



Benchmarking Aligned Reasoning in Test-Time Constrained AI Systems

RAPHAEL YANA

UNIVERSITY COLLEGE LONDON
COMPUTER SCIENCE DEPARTMENT

Submitted to University College London (UCL) in partial
fulfilment of the requirements for the award of the degree of
Master of Science.

Primary supervisor: **PHILIP TRELEAVEN**

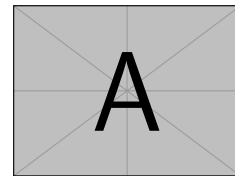
External supervisor: **ADRIANO KOSHIYAMA**

Word count:

Thesis submission date: **22/09/2025**

Declaration

I, **RAPHAEL YANA**, confirm that the work presented in this thesis is my own. Where information has been derived from other sources, I confirm that this has been indicated in the thesis.



RAPHAEL YANA
London, United Kingdom
22/09/2025

Abstract

This thesis develops a comprehensive framework for evaluating and improving the robustness of large language models (LLMs) in reasoning tasks. As LLMs become integral to high-stakes applications from content moderation to educational assistance understanding their behavior under normative stress becomes critical. Current evaluation practices, however, predominantly focus on single-metric accuracy assessments that fail to capture the nuanced ways models handle ambiguity, respond to demographic framing, or maintain consistency across similar cases. This work addresses these gaps through a systematic investigation combining cost-aware benchmarking, demographic profile analysis and consensus-based stability metrics.

Study 1: Benchmarking with budget constraints. The research unfolds across three interconnected empirical studies. The first study establishes a cost-aware benchmark comparing six in-context learning strategies across normative tasks (MGSD for stereotype detection, MentalManip for manipulation detection) and general reasoning (MMLU). Results reveal that few-shot prompting with concise definitions achieves optimal accuracy-cost trade-offs, while complex multi-stage reasoning strategies increase computational overhead without proportional performance gains. Among tested models, GPT-4.1-mini demonstrates superior accuracy and stability within practical budget constraints.

Study 2: Profile inference and consensus selection. The second study investigates how demographic role-playing influences model reasoning through systematic evaluation of 60 profiles varying across gender, ethnicity, and age dimensions. While descriptive analysis reveals small performance gaps (typically below 3%) and generalized linear models show non-significant effects on most datasets, permutation tests uncover subtle non-linear patterns in MGSD and MMLU-Large. Critically, these demographic effects remain operationally minor, with task and item difficulty emerging as the primary performance drivers rather than demographic framing.

Study 3: Stability of profile effects. The third study examines the stability of profile-induced behaviors by developing novel metrics linking cross-validated consistency to boldness measures based on leave-one-out consensus. Analysis reveals negative correlations between consistency and boldness on MMLU and MGSD, with boldness showing modest positive correlation with accuracy only on MMLU. Rescue-versus-error analysis demonstrates that inconsistent profiles provide marginal rescue advantages on MMLU but not other datasets. Predictive attribution confirms that while demographic factors explain some in-sample

variance, out-of-sample generalization remains weak, suggesting profile effects are largely dataset-specific rather than systematic.

This thesis makes three principal contributions to the field. First, it provides a state-of-the-art, cost-aware benchmark of in-context learning strategies across normative and objective tasks, offering practical guidance for deployment decisions. Second, it introduces a multi-level analytical framework capable of detecting demographic disparities, instability patterns, and profile-induced behavioral shifts through complementary statistical approaches. Third, it develops a consensus-based methodology that rigorously connects consistency, boldness, and benefit-risk trade-offs, providing empirical evidence for when profile diversity enhances performance versus when simple few-shot prompting suffices. Together, these contributions advance our understanding of LLM robustness in morally complex contexts and provide actionable insights for developing more reliable AI systems.

Acknowledgements

I would like to address many thanks to my supervisor, Dr. Philip Treleaven, for the dedicated amount of time in this project. I would also like to address my thanks to Holistic AI's members for their guidance. Without all this help, this project would not have gone this far.

Contents

1	Introduction	8
1.1	Motivation	8
1.2	Thesis Overview	9
1.3	Research Objectives and Contributions	10
1.4	Thesis Structure	10
2	Background and Literature Review	12
2.1	Background Theory	12
2.1.1	Notations	12
2.1.2	Normative Reasoning in Large Language Models	13
2.1.3	Test-Time Constraints	14
2.1.4	In-Context Learning Strategies	14
2.1.5	Evaluation under Uncertainty	15
2.2	Related Work	17
2.2.1	Normative Reasoning Datasets	17
2.2.2	In-Context Learning and Test-Time Reasoning Strategies	19
2.2.3	Role-Playing Effects in LLMs	21
2.2.4	Bias, Fairness and Demographic Disparities Evaluation	23
2.2.5	Stability and Uncertainty	26
2.2.6	Consensus Methods Under Token Budgets	28
2.2.7	Research Gaps	29
3	Benchmarking with Budget Constraints	32
3.1	Methodology	32
3.1.1	Overview and Research Questions	32
3.1.2	Experimental Design	32
3.2	Results	34
3.2.1	Strategy Performance	35
3.2.2	Model Performance	37
3.2.3	Dataset Patterns	37
3.3	Discussion	37
3.3.1	Selection Guidelines	37
3.3.2	Dataset Characteristics	38
3.3.3	Limitations	38
4	Profile Inference and Consensus Selection	40
4.1	Methodology	40
4.1.1	Data, models and strategies	40
4.1.2	Level 1: Descriptive Analysis	41

4.1.3	Level 2: Confirmatory Inference	42
4.1.4	Level 3: Nonparametric Verification	43
4.2	Results	43
4.3	Discussion	46
5	Stability of Profile Effects	50
5.1	Methodology	50
5.1.1	Data structure, baseline, and outcomes	50
5.1.2	Ambiguity and boldness metrics	50
5.1.3	Consistency across Folds	51
5.1.4	Statistical Analysis	51
5.2	Predictive Attribution Modeling	52
5.2.1	Normative assessment of boldness via nonparametric tests	52
5.3	Results	53
5.4	Discussion	56
6	Discussion and Conclusion	60
6.1	Conclusion	60
6.1.1	Research Implications	62
6.1.2	Limitations and Future Directions	63
A	Resources	65
A.1	Code	65
B	Study 2: Additional Contents	66
B.1	Empirical Group-Level Accuracy	66
B.2	Error Directionality	66
B.3	High-disagreement evaluation	68
B.4	Additional Results	68

Chapter 1

Introduction

1.1 Motivation

The deployment of large language models in real-world applications has accelerated dramatically, extending beyond traditional natural language processing tasks into domains requiring nuanced moral and social reasoning. LLMs such as GPT-4, Claude, and Gemini are increasingly embedded in systems where decisions carry ethical weight: educational platforms that shape young minds, content moderation systems that define acceptable discourse, healthcare interfaces that influence medical decisions, and legal assistants that affect justice outcomes. In these contexts, model failures transcend mere inaccuracy; they risk amplifying societal biases, enabling manipulation, and producing inconsistent moral judgments that undermine trust in AI systems.

Despite impressive performance on standard benchmarks, current evidence suggests that LLMs remain fragile under what we term "normative load" situations where reasoning tasks involve moral, ethical, or socially charged dimensions. This fragility manifests in three critical ways. First, these models struggle with inherent ambiguity in normative questions where multiple valid perspectives exist and outcomes depend heavily on ethical framing. Second, they demonstrate vulnerability to adversarial manipulation, where carefully crafted inputs can elicit unsafe, biased, or contradictory outputs. Third, they exhibit systematic inconsistencies when prompted from different demographic or ideological perspectives, raising fundamental questions about fairness and reliability.

Current evaluation paradigms inadequately address these challenges. The prevailing focus on aggregate accuracy metrics obscures the mechanisms underlying model decisions and fails to capture consistency across similar cases. Most evaluations adopt a single, ostensibly neutral perspective, overlooking how dramatically LLM behavior can shift when prompted to adopt different demographic profiles or cognitive styles. Furthermore, the reliance on isolated datasets or narrow task domains limits our ability to assess whether observed patterns generalize across varying levels of normative complexity.

These limitations underscore the urgent need for more sophisticated evaluation frameworks that can systematically compare in-context learning strategies under different types of normative stress, diagnose

how perspective shifts influence reasoning patterns, and test whether aggregating diverse viewpoints can yield more robust and equitable outputs. This thesis addresses this need through an integrated framework that advances both diagnostic capabilities and practical mitigation strategies for improving LLM reliability in morally complex contexts.

1.2 Thesis Overview

This thesis presents a three-study empirical analysis that progressively builds from foundational benchmarking through diagnostic analysis to stability assessment. Each stage addresses distinct aspects of the research problem while contributing to a comprehensive understanding of LLM behavior under normative load.

Stage 1: Cost-Aware Benchmarking. We establish baseline performance by systematically comparing six in-context learning strategies: zero-shot, few-shot, definition-augmented, Chain-of-Thought, Tree-of-Thought, and generated-knowledge approaches across three complementary datasets. MGSD tests stereotype detection capabilities, MentalManip evaluates manipulation resistance, and two MMLU variants assess general reasoning. By measuring both accuracy and computational efficiency, we identify pragmatic trade-offs between performance and resource consumption, providing actionable guidance for deployment decisions.

Stage 2: Profile Inference and Consensus . We conduct systematic evaluation across 60 demographic profiles varying gender, ethnicity, and age to understand how role-playing affects model reasoning. Our analysis proceeds through three complementary levels: Level 1 provides descriptive statistics on performance gaps and error patterns; Level 2 applies generalized linear models to test for systematic effects; Level 3 employs non-parametric permutation tests to detect subtle, potentially non-linear relationships. This multi-pronged approach ensures we capture both obvious disparities and nuanced behavioral shifts.

Stage 3: Stability of Profile Effects. We develop novel metrics to quantify the relationship between consistency and decision-making boldness. Through cross-validated consistency measures and consensus-based boldness metrics derived from leave-one-out analysis, we examine how different profiles balance reliability against willingness to make difficult judgments. We investigate correlations between consistency, boldness, accuracy, rescue rates, and extra errors, while attributing variance sources and assessing out-of-sample generalization potential.

1.3 Research Objectives and Contributions

The primary objective of this research is to develop and validate methods for evaluating and improving LLM reliability in tasks involving varying degrees of normative complexity and perspective sensitivity. This overarching goal encompasses several specific objectives:

- To establish comprehensive benchmarks comparing in-context learning strategies across normative and general reasoning tasks while accounting for computational costs
- To quantify how demographic role-playing influences model behavior through rigorous multi-level statistical analysis
- To develop metrics that capture the trade-offs between consistency and boldness in model decision-making
- To provide empirical evidence guiding the deployment of profile-based versus standard in-context learning approaches

In pursuing these objectives, this thesis makes three principal contributions to the field:

Contribution 1: Cost-Aware Benchmarking Framework. We provide the first systematic comparison of in-context learning strategies that explicitly considers the accuracy-efficiency trade-off across normative and objective tasks. Our results offer practical guidance for selecting in-context learning strategies based on task requirements and resource constraints.

Contribution 2: Multi-Level Demographic Analysis Methodology. We introduce a comprehensive framework for detecting and quantifying demographic disparities in LLM reasoning through complementary statistical approaches. This methodology reveals both linear and non-linear effects while distinguishing operationally significant patterns from statistical artifacts.

Contribution 3: Consensus-Based Stability Metrics. We develop novel metrics linking consistency to boldness through leave-one-out consensus analysis, providing new tools for understanding benefit-risk trade-offs in model behavior. Our framework offers empirical criteria for determining when profile diversity enhances performance versus when simpler approaches suffice.

1.4 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 reviews relevant literature on in-context learning strategies, bias evaluation, normative benchmarks, and profile-based analysis, situating our work within the broader research landscape. Chapter 3 presents the cost-aware benchmarking study, establishing baseline performance across in-context learning strategies and identifying optimal accuracy-efficiency

trade-offs. Chapter 4 details the multi-level profile analysis, mapping systematic patterns in demographic-influenced reasoning. Chapter 5 develops the stability and boldness framework, examining consistency-boldness relationships and their implications for deployment. Chapter 6 synthesizes findings, discusses implications for AI safety and deployment, and outlines directions for future research in normative robustness evaluation.

Chapter 2

Background and Literature Review

This chapter introduces theoretical concepts and surveys recent research on large language models evaluation on normative reasoning tasks under resource constraints.

2.1 Background Theory

2.1.1 Notations

This thesis focuses on inference-time behavior that can be modified without model re-training, focusing on controlled comparisons in normative reasoning tasks with role-playing and resource constraints.

Sets We define $\mathcal{I}$ as the set of evaluated samples (or items), with $n = |\mathcal{I}|$. We denote the label space $\mathcal{K}$ (*e.g.* $\{0; 1\}$ for MentalManip and MGSD). We also define the sets of models $\mathcal{M}$ and strategies $\mathcal{S}$ (zero-shot, few-shot, etc.). For role-playing, we must define the set of studied profiles $\mathcal{P}$. We define each profile's trait t (*e.g.* ethnicity, gender) with $g \in G_t$ (*e.g.* men, women), where $\mathcal{P}_g$ is the set of profiles carrying trait g . We also define a neutral baseline $p_0 \in \mathcal{P}$ used for comparison.

Predictions For a pair of given sample $i \in \mathcal{I}$ and profile $p \in \mathcal{P}$, we record the predicted label $\hat{y}_{i,m,s,p} \in \mathcal{K}$ and the ground truth $y_i \in \mathcal{K}$ where $m \in \mathcal{M}$, $s \in \mathcal{S}$. We define the corresponding indicator function¹ [1] for prediction's correctness:

$$Y_{i,m,s,p} = \mathbf{1}[\hat{y}_{i,m,s,p} = y_i] \in \{0, 1\}$$

Metrics for Classification In binary settings, by considering 1 as the "positive" label, we can define four metrics: *true positive TP*, *true negative TN*, *false positive FP* and *false negative FN*.

We compute the **accuracy** $acc_{m,s,p}$ per model-strategy-profile combination [2]:

$$acc_{m,s,p} = \frac{1}{n} \sum_{i \in \mathcal{I}} Y_{i,m,s,p} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2.1.1)$$

¹ $\mathbf{1}[X] = 1$ if X is True, 0 if X is false.

representing the probability of the model making a correct prediction.

We also define the **per-trait mean accuracy** $\bar{a}_{m,s}(g)$ as:

$$\bar{a}_{m,s}(g) = \frac{1}{|\mathcal{P}_g|} \sum_{p \in \mathcal{P}_g} acc_{m,s,p} \quad (2.1.2)$$

where profiles are considered as exchangeable samples of trait g .

We then define the **precision** as:

$$Precision = \frac{TP}{TP + FP} \quad (2.1.3)$$

representing how confident we can be about a positive prediction given a sample [2].

The **recall** is defined as:

$$Recall = \frac{TP}{TP + FN} \quad (2.1.4)$$

representing the ability of the model to extract positive labels from the dataset [2].

We finally define the **F1-score**, a common metric to assess classification model's performance [2]:

$$F1-Score = 2 \left(\frac{Precision \times Recall}{Precision + Recall} \right) \quad (2.1.5)$$

We must note accuracy can be applied to any classification task, while *precision*, *recall* and *F1-score* cannot be applied in multi-class settings (*e.g.* MMLU).

2.1.2 Normative Reasoning in Large Language Models

Normative reasoning comprises tasks on evaluating logical relationships among propositions expressing directives or value judgments [3]. In contrast to factual reasoning where an objective ground truth exists, normative domains require models to operate in contexts where several interpretations coexist. This ambiguity leads to unique challenges for evaluation, as model's response may be incorrect because it applies a valid normative framework (or perspective) different to the one assumed by evaluators. Therefore, errors do not occur because a model fails to reason correctly.

This challenge intensifies when combining normative judgments with cultural variation. Ethical behavior or moral reasoning foundations vary across communities or genders, leading to implicit difficulties in benchmark design. As models are trained on a large corpus of diverse data, they may also internalize conflicting normative frameworks and their responses may shift due to subtle prompt variations or demographic role-playing. This constitutes a feature of normative reasoning which increases difficulty of evaluation when benchmarks assume a single correct answers.

2.1.3 Test-Time Constraints

Deployment of AI systems requires strict resource budgets which affect strategies performance.

Defining the budget B_s as the maximum allowed tokens during evaluation for a given strategy $s \in \mathcal{S}$ [4, 5], we discern two main token consumption values, the prompt tokens $T_{i,m,s,p}^{\text{prompt}}$ and the completion tokens $T_{i,m,s,p}^{\text{completion}}$. The total consumed tokens per item is defined as [4]:

$$T_{i,m,s,p} = T_{i,m,s,p}^{\text{prompt}} + T_{i,m,s,p}^{\text{completion}} \quad (2.1.6)$$

Out of these metrics, we can define the average consumption over items similarly to [4, 6, 7]:

$$\bar{T} = \frac{1}{n} \sum_{i \in \mathcal{I}} T_{i,m,s,p} \quad (2.1.7)$$

and the accuracy per 1 000 tokens acc_{PT} :

$$acc_{PT} = \frac{acc_{m,s,p}}{\frac{T_{m,s,p}}{1000}} = \frac{1000 \times acc_{m,s,p}}{T_{m,s,p}} \quad (2.1.8)$$

and the tokens required per correct answer $T_{\text{correct},m,s,p}$:

$$T_{\text{correct},m,s,p} = \frac{\bar{T}_{m,s,p}}{acc_{m,s,p}} \quad (2.1.9)$$

permitting to assess and compare multiple strategies with different token consumption rates based on their efficiency.

2.1.4 In-Context Learning Strategies

Throughout this thesis, we evaluate five in-context learning strategies widely used in the field, with distinct required budgets and effects: Zero-shot (ZS), Few-shot (FS), Chain-of-Thought (CoT), Tree-of-Thought (ToT) and Generated Knowledge Prompting (GKP).

Zero-shot This strategy introduced first by Wei et al. [8] is a form of instruction learning. By providing the instructions and the question, we perform a sort of fine-tuning permitting the model to answer. It also permits to test models' ability to recall pre-trained knowledge without context.

Few-shot This strategy augments the zero-shot prompt with examples (samples from evaluated dataset) [9]. These examples are formatted to show both the input and the correct label to the LLM. This permits to test whether models can learn and apply patterns from demonstrations [10].

Chain-of-Thought The Chain-of-Thought (CoT) strategy requests to the model a step-by-step reasoning before answering. This strategy permits to test whether guided and explicit reasoning improves accuracy on the task despite an increased consumption.

Tree-of-Thought The Tree-of-Thought (ToT) strategy extends CoT through multiple reasoning paths. This strategy tests whether diversity in reasoning improves the accuracy and if LLMs are able through this reasoning to emit hypothesis they could believe in.

Generated Knowledge Prompting Generated Knowledge Prompting (GKP) strategy performs a two-stage inference: firstly generating the relevant background theory from pre-trained knowledge, then answering using both the original question and newly generated context as support. This strategy tests if models can self-augment their context to improve reasoning effectively.

LLM-as-a-Judge This strategy consists in using LLMs to evaluate outputs from other AI systems. It leverages a model’s capacity to judge, with a potential for efficiency and comparable results to human raters in some benchmarks [11] [12]. However, they remain subject to inconsistencies, biases and manipulation by adversarial prompt [13].

Since minor structural changes can result in large accuracy variations due to prompt sensitivity [14], each of these strategies faces challenges. Definition injection is a potential solution to reduce variance [15, 16, 17].

2.1.5 Evaluation under Uncertainty

Quantifying Uncertainty

Under role-playing settings, given $\mathcal{P}_g \subset \mathcal{P}$ for a trait g , we define the empirical accuracy per profile:

$$\hat{p}_{g,p} = \frac{1}{n} \sum_{i \in \mathcal{I}} Y_{i,p} \quad p \in \mathcal{P}_g \quad (2.1.10)$$

We treat the binary outcomes of the classification as:

$$Y_{i,p} \sim \text{Binom}(n, \hat{p}_{g,p}) \quad (2.1.11)$$

where:

$$\hat{p}_{g,p} = \frac{X}{n}$$

Thus, we can associate to this empirical accuracy its $(1 - \alpha)$ Wilson binomial interval $\text{CI}_{\text{Wilson}}$ for quantifying the binomial uncertainty within the profile itself, defined as:

$$\text{CI}_{\text{Wilson}}(\hat{p}_{g,p}) = \frac{\hat{p}_{g,p} + \frac{z^2}{2n}}{1 + \frac{z^2}{n}} \pm \frac{z}{1 + \frac{z^2}{n}} \sqrt{\frac{\hat{p}_{g,p}(1 - \hat{p}_{g,p})}{n} + \frac{z^2}{4n^2}} \quad (2.1.12)$$

where $z = z_{1-\alpha/2}$ is the $1 - \frac{\alpha}{2}$ quantile of the standard normal distribution. This confidence interval permits to ascertain confidence on the empirical accuracy if we increased the number of samples studied.

We interpret the selected profiles given a trait group as exchangeable samples from a larger population of profiles exhibiting trait g . Therefore, we summarize a trait level by the mean across its profiles:

$$\bar{p}_g = \frac{1}{m_g} \sum_{p \in \mathcal{P}_g} \hat{p}_{g,p} \quad (2.1.13)$$

where $m_g = |\mathcal{P}_g|$ to which we associate the across-profile t -interval:

$$\bar{p}_g \pm t_{1-\alpha/2, m_g-1} \frac{s_g}{\sqrt{m_g}} \quad (2.1.14)$$

And:

$$s_g^2 = \frac{1}{m_g - 1} \sum_{p \in \mathcal{P}_g} (\hat{p}_{g,p} - \bar{p}_g)^2 \quad (2.1.15)$$

The associated standard error $\text{SE}(\bar{p}_g) = s_g/\sqrt{m_g}$ captures the variability between profiles² within a trait g .

Eq. (2.1.14) and (2.1.15) are undefined when $m_g = 1$, *i.e.* when a trait group contains only one profile.

Statistical Tools

For this thesis, we must define some other statistical tools, used later in the studies: *McNemar's test*, the *exact Binomial test*, *odds ratio*, *Wald confidence interval* and the *Haldane-Anscombe correction*.

We then define the McNemar test, a widely used test for classification under binary outcomes. It permits to evaluate whether two classifiers or groups perform similarly on a task, based on their predictions but ignoring their correctness on n items. For two profiles p_1 and p_2 , we define:

$$b = \sum_{i \in \mathcal{I}} \mathbf{1}\{Y_{i,p_1} = 1, Y_{i,p_2} = 0\}$$

and:

$$c = \sum_{i \in \mathcal{I}} \mathbf{1}\{Y_{i,p_1} = 0, Y_{i,p_2} = 1\}$$

The tested hypothesis in McNemar's test is:

$$H_0 : P(Y_{i,p_1} = 1, Y_{i,p_2} = 0) = P(Y_{i,p_1} = 0, Y_{i,p_2} = 1) \quad (2.1.16)$$

Therefore, McNemar's test is defined as:

$$\chi^2 = \frac{(b - c)^2}{b + c} \quad (2.1.17)$$

For small testing samples, unbalanced categories or when $b + c < 25$, the Fischer's exact test is preferable. Under H_0 , we assume both group

²It differs from the binomial uncertainty within a profile $\text{CI}_{\text{Wilson}}$ associated to a single $\hat{p}_{g,p}$.

perform similarly and thus, the disagreement pairs follow a binomial distribution. The exact p -value for H_0 is:

$$p_{\text{exact}} = 2 \min \left(\sum_{k=0}^{\min(b,c)} \binom{b+c}{k} 0.5^{b+c}, \sum_{k=\max(b,c)}^{b+c} \binom{b+c}{k} 0.5^{b+c} \right) \quad (2.1.18)$$

An important metric related to this is the odd's ratio of disagreement pairs, defined as:

$$\theta = \frac{b}{c} \quad (2.1.19)$$

for which $\theta = 1$ indicates perfect symmetry of the two groups performances.

We can define the Wald confidence interval representing the log-normal 95% confidence interval for the odds ratio:

$$\text{CI}_{\text{Wald}} = \exp \left\{ \log \theta \pm 1.96 \sqrt{\frac{1}{b} + \frac{1}{c}} \right\} \quad (2.1.20)$$

But as this confidence interval is undefined when $b = 0$ or $c = 0$ since the odds ratio and confidence intervals will be equal to $+\infty$, the Haldane-Anscombe correction permits to measure the interval even in those extreme cases by adding 0.5 to each cell of the contingency table:

$$\theta_{\text{corrected}} = \frac{b + 0.5}{c + 0.5} \quad (2.1.21)$$

leading to a corrected confidence interval:

$$\text{CI}_{\text{Wald,corrected}} = \exp \left\{ \log \theta_{\text{corrected}} \pm 1.96 \sqrt{\frac{1}{b + 0.5} + \frac{1}{c + 0.5}} \right\} \quad (2.1.22)$$

2.2 Related Work

The literature on LLM evaluation for normative reasoning has evolved along several paths. This review is organized into six axes, addressing different aspects of the problem while contributing to our understanding of how to achieve fair, efficient and reliable assessment under resource constraints.

2.2.1 Normative Reasoning Datasets

Large language models (LLMs) evaluation in normative reasoning tasks has emerged as a critical area of research, particularly as these models are increasingly deployed in sensitive contexts where we require ethical judgment and fairness considerations. This review inspects the current state of benchmarking methodologies for LLMs under fixed test-time scale constraints, with a focus to fairness metrics and role-playing effects which may introduce systematic biases into evaluations. While resource constraints and fairness requirements intersect, unique challenges

rise: models must perform accurately and equitably across diverse demographic groups while being limited by strict resource limitations to reflect real-world deployment scenarios.

Recent surveys established foundational taxonomies for bias and fairness understanding in LLMs. Gallegos et al. [18] provide a formal standardization of bias evaluation frameworks, by defining critical metrics that supports fairness analysis across both classification and generative tasks. They identify three primary bias categories: representational biases that affect the different groups depictions in model outputs, allocation biases that result in varying performance across demographics and quality-of-service biases that manifest as varying reliability for different populations. While this work provides essential definitions on social bias evaluation criteria and operational fairness requirements, it offers limited guidance on how these biases interact under resource limitations. This represents a critical gap as deployment must balance fairness and computational efficiency.

Gupta et al. [19] demonstrate empirically that persona role-playing triggers statistically significant performance differences across protected groups, with reasoning biases emerging consistently across several domains. They observe that large variations of performance occur in normative reasoning tasks and these effects are intensified for historically marginalized groups. Though they detect a 40% accuracy variation, their experiments' reliance on explicit demographic traits may overestimate real situations bias effects, as demographic information is often implicit and contextual rather than explicitly stated.

This review connects five directions of research that collectively support a multiple-level evaluation framework. Firstly, we inspect the primary datasets used for normative reasoning assessment, focusing on design validity and contamination risks that can threaten benchmark results integrity. Secondly, we analyze in-context learning strategies and their impact on test-time reasoning under resource budget constraints, revealing the link between mitigation or intensification of fairness issues and prompt engineering approaches. Thirdly, we investigate bias and fairness considerations in persona-based evaluations, documenting systematic disparities which emerge when models are evaluated under different demographic descriptions. Fourthly, we assess stability and reproducibility challenges in current benchmarking methods for LLMs and variance source identification associated to fairness problems. Finally, we explore consensus methods to optimize performance within token budgets while insuring fairness, illustrating how targeted consensus methods can improve both accuracy and equity.

These directions converge to support a hierarchical evaluation framework divided in four levels, aligned with the mixed benchmark under controlled test-time budget studied in this thesis. Level 1 conducts descriptive assessment of accuracy, profile disagreement and raw group gaps, followed by confirmatory inference in Level 2 on clustered logistic models for traits main effects. Level 3 follows by providing consensus recommendations and corroborates disparities with permutation tests. Level 4 finally provides a stability exploration with cross-fold consistency and variance attribution to demographic traits. Each level is connected in

a specific order, resulting in a comprehensive framework which acknowledges the complex nature of fair and reliable LLM evaluation. This structure underlines the understanding of fairness in the context of resource constraints, prompt sensitivity and evaluation stability. These dataset limitations directly motivate the need for in-context learning strategies with ability to maintain robust performance even on imperfect targets, as we explore it in the following section.

2.2.2 In-Context Learning and Test-Time Reasoning Strategies

Given the limitations identified above in datasets, effective evaluation requires robust in-context learning strategies to these underlying data quality issues. Reliable LLM evaluation depends on high-quality datasets that test reasoning capabilities accurately by measuring the intended skill, while avoiding bias contamination. This work focuses on three primary benchmarks, each with distinct strengths and documented limitations that shape our understanding of model capabilities and underlying biases.

The MMLU benchmark was created to address contamination concerns and achieve objectivity in answers. However, Zhao et al. [20] reveal that despite this intent, the original MMLU implementation suffers from significant contamination issues. Their research shows that traditional MMLU implementations could overestimate model capabilities by 15-20% due to shared training data, with contamination rates varying highly across subjects. The humanities and social sciences questions exhibit contamination rates exceeding 30%, while these rates remain lower but still substantial around 12% for STEM fields. This differential contamination creates artificial performance gaps which can be mistaken for true capability differences. To address these issues, Zhao et al. [20] introduce variants specifically designed to limit data leakage and improve evaluation quality. Their approach consists in implementing a three-stage decontamination framework: sentence matching to identify exact duplicates, semantic similarity detection using embedding models for paraphrase interception and adversarial probing by testing model’s ability to complete partial questions from the benchmark. Their contamination-free dataset includes newly designed multiple-choice questions validated by subject experts and tested on human baselines to ensure difficulty calibration. However, Zhao et al.’s [20] decontamination approach only addresses explicit memorization. These methods do not tackle subtle contamination through shared reasoning or implicit knowledge overlap which may still increase performance estimations. Recent surveys also underline the possibility that subject composition imbalance and corpus-level alignment effects may still introduce residual errors. They motivate carefulness in evaluation set design, especially for external validity and generalization to less-represented subjects.

The Mental Manipulation (MentalManip) dataset, introduced by Wang et al. [21], evaluates models’ understanding of psychological manipulation and social dynamics. It is composed of approximately 4,000 multi-turn dialogues extracted from movie scripts. While this fictional set

provides many examples of manipulative situations, it may not fully reflect real-world communication patterns due to its scripted nature. Each dialogue is annotated for manipulative language presence in a binary classification setting. The annotation process is described as following: multiple raters followed detailed guidelines and proceed to individual votes. This subjectivity creates validity issues, since, if expert annotators cannot achieve a high agreement on manipulation identification, the benchmark may be measuring the annotators preferences rather than the true manipulative intentions in dialogue. This observation raises questions on its utility for model evaluation. Additionally, the annotator group presented demographic imbalances, mainly in gender, though inter-annotator agreement showed no significant gender-based variations. The authors acknowledge the challenge that achieving a representative annotator sample remains an ongoing challenge in dataset construction. These limitations inform directly our evaluation framework’s first level, where descriptive accuracy evaluation must report confidence interval for label uncertainty. This thesis therefore combines paired item evaluation to budget-aware designs and reports confidence intervals with BH-FDR (where possible), which treats intra-group gaps as exploratory. Mental-Manip motivates a assessment on real-world validity.

Finally, the MGSD benchmark created by Wu et al. [22], serves as resource for assessing stereotype detection capabilities in large language models. MGSD consists of over 50 000 divided in categories such as gender, race, profession and religion. It is studied in a binary classification setting in [22, 23]. Its construction, stemming from two other datasets, provides strong links to social psychology and bias detection but exhibits also limitations. Its coverage of marginalized groups is unbalanced and some dimensions, such as racial and gender minorities stereotypes, are underrepresented due to limitations in the source datasets. Additionally, generic statements can occur because of linguistic style and context which may not capture the subtlety of stereotypes in real-world scenarios. Classification implementations also focus on explicit and context-independent labeling, hardening the task for more nuanced forms of bias. These representation gaps require our framework’s second level corresponding to confirmatory inference approach, where demographic effects are tested across clustered models able to discern systematic bias from sampling artifacts.

Critical challenges in dataset construction and validity have been particularly observed in Theory of Mind (ToM) evaluations. Many recent ToM studies serve as benchmarks in both the human and LLM literature [24], by implementing false-beliefs paradigms such as Sally-Anne test (introduced by Baron-Cohen et al. [25]). Although they remain widely used, Riemer et al. [26] criticizes this category of benchmarks for measuring pattern-matching instead of in-context adaptation of reasoning in agents, where they exhibit flexible mental state and thus fail to measure psychological constructs when applied to LLMs. Their research shows that model performance remains largely unchanged when modifying belief attribution questions while preserving syntax. It suggests models rely mainly on non-causal statistical associations rather than mental state reasoning. This critique is reinforced by Wang et al. [27], who report in

their work robustness issues through adversarial testing and defend the need for tailored metrics which account for the fundamental differences between human and artificial reasoning processes. The alternative they propose includes dynamic scenarios where mental states evolve over time and cannot be inferred from surface patterns alone.

Contamination detection has risen as a methodological concern affecting all major benchmarks. The DCR framework by Xu et al. [28] provides rigorous approaches for detecting contamination through prompt modification, discovering up to 30% of benchmark performance gains may be attributable to data leakage. Their detection method combines membership inference attacks with perturbation analysis to identify instances where models have memorized data rather than learned in the context. Fu et al. [29] findings on inconsistent detection effectiveness across several methods reveals a concerning issue: the designed tools for benchmark integrity validation benchmark cannot themselves be reliably validated, possibly permitting undetected contamination in widely used evaluation frameworks. These inconsistencies across detection methods strengthen our framework’s last level for stability assessment where cross-fold validation becomes crucial for discerning contamination effects from model capabilities.

2.2.3 Role-Playing Effects in LLMs

Building on identified contamination and representation problems in evaluation datasets, maintaining evaluation validity despite these underlying data quality problems necessitate in-context learning strategies. In-context learning strategy’s choice shapes LLM performance, introducing both opportunities for enhanced reasoning and risks of systematic bias. Recent work began to structure prompt engineering implementations and their differential bias effects under several constraints, underlining complex interactions between prompt design, model architecture and task type.

Liu et al. [30] provide a detailed taxonomy identifying four main families of prompt engineering: profiles/instructions (demographic specifications, role assignments), knowledge augmentation (retrieval-augmented generation, few-shot examples, context injection), thought processes (chain-of-thought, tree-of-thought, self-consistency) and reliability improvements (constitutional AI, uncertainty expression, adversarial prompting). While Liu et al.’s work provides a useful partition, their performance observations rely on limited task sampling, which is unlikely to generalize across domains. For instance, thoughts processes improve mathematical reasoning by 35-40%, but this effectiveness vanishes (paradoxically) for basic factual tasks where such staged reasoning introduces unnecessary complexity. These findings establish that effectiveness of prompt designs varies more depending on task type than model family.

Santana et al. [31] empirically validate this taxonomy, by systematically comparing fourteen prompting techniques across multiple task setups and LLMs. Their evaluation on 50 000 prompt-model-task combinations shows that no single prompt strategy stands out across all contexts. Zero-shot prompting performs best when models have strong pre-training

on similar tasks with clear structure formats, few-shot learning outperforms for clarifying ambiguous specifications with optimal injected shots tailored to complexity of task and chain-of-thought (CoT) prompting highly improves multi-step reasoning while potentially decreasing performance on simple fact retrieval. However, Santana et al.’s [31] findings indicates a critical trade-off contradiction: ToT strategies outperform in complex reasoning tasks but their token consumption increase by a factor of 5 to 10 leading to a paradox where the most effective techniques becomes excessively expensive in real-world deployments.

Prompt sensitivity stands as a main challenge to make evaluation reproducible. Arabzadeh et al. [32] conduct comparative studies, indicating that prompt relevance and sensitivity differ between humans and LLMs. While human performance shifts less with phrasing changes (under 5%) in general, LLMs exhibit much more sensitivity: punctuation, sentence structure and synonym substitutions can cause above 20% performance variation on identical tasks. This hypersensitivity exposes a fundamental limitation in LLM evaluation: if minor structural changes can cause such high variations, the reported model capabilities may only reflect prompt engineering artifacts rather than reasoning abilities, undermining the validity of comparative studies.

Recent benchmarking work by Arora et al. [33] directly show how Pareto frontier analysis justifies the trade-offs between accuracy and resource cost for deploying LLMs in real-world situations. By empirically comparing multiple model across thousands of medical evidence summarization tasks, their research show that optimal strategies do not maximize all metrics. Instead, each model or prompt regime constitutes a distinct point on their designed performance-cost curve. Their approach reveals a clear Pareto frontier where improvements in accuracy are possible only at resources spent. The study recommends that evaluators and practitioners systematically report these trade-off frontiers to enable context-sensitive and informed selection of prompting regimes where a preferred balance between benchmark quality and efficiency can be adapted to deployment constraints. This trade-off analysis on multiple dimensions aligns with our framework’s third level (Level 3), as we ensure balance between accuracy improvements and token usage in our consensus recommendations, while maintaining fairness across demographic groups.

Security remains an increasing concern in prompt-based LLM systems. Gudino-Rosero et al. [34] analyze tasks where LLMs generate or aggregate consensus deliberation or voting and show that LLM systems are highly vulnerable to prompt injection attacks. This vulnerability is intensified in multi-agent settings where they show attack success rates exceed 70%. Their taxonomy identifies direct, indirect, composition and escape attacks. Hill et al. [35] further develop these findings, by presenting a threat model and permitting documentation real-world vulnerabilities in deployed systems. Beurer-Kellner et al. [36] synthesize six practical defense patterns for input sanitation among other methods with broadly varying efficiency depending on deployment context and threat sophistication. However, the effectiveness inconsistency of these defense patterns, mirrored by their 40-85% variation in success rates, reveals

that security considerations must be implemented throughout evaluation design rather than isolated as a separate concern addressed post-hoc.

Performance reporting methodologies have matured to address these challenges. Wang et al. [37, 38] propose latency-aware benchmark, where they introduce token efficiency and output metrics and report percentile-based latency (p50, p95, p99) over averages to assess risks. Their work demonstrates that latency variance can exceed 50% across hardware types and batch sizes. They also show improvement in one metric (accuracy, speed, resource cost) frequently implies a trade-off in another.

Jointly, these developments permit to define best practices: using pre-defined prompts tailored to token budget, testing sensitivity variants calibrated for experiments reproducibility and composite token and latency reporting to ensure fair LLM evaluation. Prompt engineering should therefore be visualized as an embedded form of hyperparameter tuning, able to overfit on benchmark if not constrained and documented.

These prompting complexities, such as artifacts or security vulnerabilities, blend the fairness challenges examined in the next section, where demographic personas meet methodological limitations resulting in systematic evaluation biases.

2.2.4 Bias, Fairness and Demographic Disparities Evaluation

The prompting vulnerabilities identified above become concerning when combined with demographic role-play, where existing fairness disparities can be highly amplified by prompt variations. Systematic biases, revealed at the intersection of persona-based evaluation and fairness assessment, challenge the validity of current benchmarking approaches. Recent research has documented how demographic role-play and persona construction introduce measurable disparities in model performance across protected groups, with implications that extend beyond accuracy metrics to justice representation and algorithmic equity.

Systematic investigation of bias presence in role-playing scenarios demonstrates consistent performance decay for marginalized demographic groups across multiple evaluation dimensions. Comprehensive research by Li et al. [39] on bias benchmarking during persona role-play reveals accuracy disparities, which can exceed 25% between demographic groups in synthetic role-play experiments, with intensified effects when intersecting multiple marginalized identities. Their analysis of 10 000 persona and task combinations shows that models with assigned personas of black women exhibit 32% lower accuracy on STEM reasoning tasks compared to personas of white men. These results are repeated even when controlling for prompt complexity, sensitivity and token count. However, Li et al.’s ablation studies do not reveal causality, only persistence. The 32% accuracy gap between the different demographic personas could reflect training data biases, model limitations or evaluation design flaws. Interpretation is ambiguous about whether observed disparities indicate concerning bias or legitimate domain knowledge differences. The work by Karnoven et al. [40] on robustly improving LLM fairness, delivers anal-

ysis on how demographic prompts interact with dataset-specific biases, finding that standard prompting approaches can amplify existing disparities in benchmarks similar to MGSD. They ascertain three main mechanisms of bias amplification: stereotype activation where demographic information triggers learned associations, contextual priming where personas influence interpretation of ambiguous questions and finally confidence modulation where models show different certainty levels based on their own perceived demographic alignment with task domains.

Persona generation aligned with population represents a crucial methodological advance, addressing limitations of ad-hoc demographic specifications. Recent frameworks by Hu et al. [41] for generating systematic persona measure bias propagation throughout prompt regimes, demonstrating that ad-hoc persona design systematically undervalues minority perspectives and ultimately reinforcing majority assumptions. Their methodology leverages large scale demographic datasets and demographic surveys to design representative persona distributions which reflect population diversity ground-truth rather than researcher intuitions. This approach reveals previously hidden biases, as models show 18% performance degradation on personas representing rural populations and 23% degradation for personas exhibiting disabilities, disparities that were previously invisible when using samples of demographic categories. The framework also introduces intersectional persona templates which capture the complexity of overlapping identities. They show that simply adding models of bias significantly lowers estimation of biases effects for individuals at the intersection of several marginalized groups.

The role-play paradox identified in recent work by Zhao et al. [42] shows that models can simultaneously present enhanced reasoning capabilities and increased errors based on stereotypes when assigned to demographic personas. This paradox underlines a dual mechanism: personas activate relevant domain knowledge which can improve performance on aligned tasks (e.g., elderly showing improved performance on historical knowledge questions) while triggering simultaneously stereotypical associations damaging performance on counter-stereotypical tasks (e.g., elderly personas showing decreased performance on questions related to technology). The assessed effect sizes are substantial, as stereotype-aligned tasks show 15-20% performance improvement while counter-stereotypical tasks show 25-30% performance decrease, resulting in a negative effect on fairness even when average performance appears stable. This paradox is intensified in dialogues with imbalanced power, as shown in research by Tan et al. [43], where demographic based accuracy changes follow predictable patterns aligned with social ranking. Models assigned what can be considered as high-status persona (CEOs, doctors, professors) demonstrate 20% better performance on leadership and decision-making tasks, while low-status personas (service workers, unemployed individuals) in contrast systematically underperform even on logical reasoning tasks independent of status.

Theory of Mind benchmarks reveal additional layers of fairness challenges, which extend beyond accuracy metrics. Comprehensive multi-modal benchmarking Villa-Cueva et al. [44] includes demographic analysis across role-playing and reasoning tasks, finding systematic disparities

in performance which correlates with demographic stereotypes embedded in training data. Their analysis of 15 000 ToM scenarios reveals models are 40% more likely to correctly attribute complex mental states to personas matching majority demographics, while showing systematic failures in understanding marginalized groups perspectives. Critical assessment of ToM benchmarks by Wang et al. [27] shows demographic variation in these benchmarks may measure social bias instead of reasoning capability. Through counterfactual interventions where demographic indicators are systematically interchanged, they show that performance differences track social stereotypes rather than logical reasoning patterns. Explicit personas necessity remains contested for instance by Lu et al. [45] research, with evidence suggesting that implicit demographic labels embedded in names, linguistic patterns and context specifications may be sufficient to trigger biased behavior even in neutral evaluations.

Recent survey work strengthens these findings across multiple evaluation frameworks, resulting in systematic patterns exceeding individual benchmarks. The comprehensive benchmarks survey by Li et al. [46] documents persona and role-play fairness diagnostic practices across 50 major benchmarks. They demonstrate 80% of these benchmarks show significant demographic disparities and 60% show disparities exceeding 15% between best and worst-performing demographic groups. Li et al.’s [46] findings that benchmarks focused on fairness still exhibit significant bias leads to a paradox. Current evaluation approaches may be fundamentally ineffective rather than improvable if benchmarks specifically designed to minimize demographic disparities cannot achieve their goal. Comprehensive reviews by Ni et al. [47] position demographic bias as a core challenge in benchmark validity and argue that current evaluation practices may be measuring models’ ability to replicate social biases rather than their reasoning capabilities. The evaluation and benchmarking survey by Mohammadi et al. [48] provides taxonomic structure for understanding how agentic frameworks link to demographic bias, identifying feedback loops where biased model outputs influence subsequent reasoning steps, amplifying initial disparities.

Uncertainty quantification in the scope of fairness assessment has emerged as a crucial methodology for understanding the reliability of bias measurements. Recent work by Ye et al. [49] on benchmarking via uncertainty quantification includes demographic fairness as an explicit statistical focus, leveraging Bayesian methods to estimate uncertainty in bias measurements and showing that a large portion of observed disparities fall within uncertainty bounds when properly quantified. This uncertainty quantification approach directly supports our framework’s confirmatory inference methodology implemented in Level 2, where Bayesian confidence intervals can isolate statistically robust disparities from measurement noise. Studies of judgments with no gold standard such as research by Xu et al. [50] find role-playing based demographic error structures remain even in unsupervised setups where no ground truth exists, suggesting that biases are deeply embedded in model representations rather than arising from biased labels, *i.e.* data contamination in labelling process. The STEM framework introduced by Hu et al. [51] provides best practices for explicitly assessing demographic bias in per-

formance evaluations, introducing relative comparison metrics which are robust to absolute performance differences while remaining sensitive to fairness violations.

These developments support the design implication that designed trait sets with comprehensive reporting of error corrections compared to extra-errors introduced in classification by persona traits and categories, including confidence intervals, are essential to assess fairness meaningfully. These findings necessitate our first level’s descriptive assessment approach requiring integrated fairness metrics from the outset of any evaluation framework, as we mitigate the fact raw accuracy metrics alone cannot capture the complex intersection of demographic bias or prompt sensitivity.

2.2.5 Stability and Uncertainty

The method vulnerabilities and demographic biases identified above can raise essential concerns when joined to the systematic instability in current evaluation practices, as measurement noise can amplify fairness issues. The reproducibility challenge in machine learning has risen in LLM benchmarking where instability in evaluation compromise confidence in reported performance gains and increase difficulty in identifying if observations represent advances or statistical noise. Recent research has documented sources of instability and proposed methodological solutions which focus on delivering robust and replicable assessment while acknowledging the challenges of evaluating systems with billions of parameters and their complex emerging behaviors.

Evidence of systematic instability spreads at every level of evaluation in current benchmarking practices. The PromptSE framework introduced by Ma et al. [52] evaluates prompt stability through controlled variations and cross-model aggregation, revealing that performance can vary by increasing up to 30% across semantically equivalent designed prompts. Their analysis includes 1000 prompt paraphrases across 20 different tasks and demonstrates that differences in prompt sensitivity varies across model families: GPT-series models show 15-20% variation in results, while open-source models such as LLaMA provide 25-35% variations in results. These results suggest architectural differences in robustness. Through factor analysis, they identify main causes of instability which include grammatical structure variations, lexical choices and punctuation patterns that should remain semantically neutral to the models but result in triggering different model behaviors. Research by Errica et al. [53] provide quantification of model sensitivity to prompt and input variation, leveraging comprehensive error attribution analysis. Their framework decomposes total error, revealing only 15% of observed errors arise from true model limitations, while 85% originate from evaluation artifacts. This raises the concern that most published LLM performance comparisons may be measuring experimental design quality rather than model capabilities.

Structural and positional variation in prompts introduce additional layers of biases and instability which jointly increase prompt sensitivity effects. Research by Cobbina et al. [54] on positional bias in test-time

prompting leverages fixed seed together with strict cross-fold replication practices, revealing that demonstration position can alter performance by 15-20% with induced effects varying considerably across positions. Early positions in prompts show underweighting biases (*i.e.* the model can forget information easily as the query appear), middle positions suffer from attention decay (models ignore easily the information) and late positions in prompts benefit from improvement in answer accuracy. These permutation protocols directly inform our framework fourth level’s cross-fold consistency approach, where systematic position effects must be controlled to accurately decompose and associate performance variance to demographic traits rather than evaluation artifacts.

The fundamental challenge to benchmark reliability is depicted in recent work by Vendrow et al. [55], arguing that label noise mainly controls the observed instability in evaluation benchmarks. Through resampling experiments on major benchmarks, they demonstrate that 30-40% of performance variance can be associated to label noise rather than behavior of models. Their introduced concepts of split-half reliability testing, paired estimate protocols and nonparametric methods provide rigorous tools for discerning signal from noise in benchmark results. However, Vendrow et al.’s [55] split-half results lead to methodological concerns, as we can wonder if models showing significant improvements on full benchmarks fail to maintain these gains on random halves, could research in the field be systematically overestimating progress due to evaluation overfitting rather than confirmed improvements in research.

Label noise represents a harming source of instability which interacts with other variance sources in complex ways. The NOISEBENCH framework introduced by Cobbina et al. [56] depicts empirical evidence for model performance degradation under real compared to artificial label noise, with real-world noise patterns showing clustered errors and creating systematic biases rather than random degradation of performances. Their analysis of 10 annotations reveals inter-annotator agreement rates below 70% for complex normative reasoning tasks with disagreements often reflecting legitimate interpretive (*i.e.* norm) differences rather than errors. Recent work by Hasan et al. [57] introduce the UPLME framework integrating Bayesian neural networks with crowdsourcing to quantify the impact of noisy labels. This multiplicative effect of instability reveals a concerning cycle. Prompt sensitivity is easily increased by noisy training data, which amplifies evaluation variance. This outcome intensifies the challenge for discerning model capabilities from evaluation artifacts, underlining a necessity for more robust statistical frameworks in LLM assessment.

Recent comprehensive survey by Devic et al. [58] reviews forty uncertainty quantification methods for large language models and categorizes them into aleatoric methods capturing data uncertainty and epistemic methods capturing model uncertainty. Their empirical comparisons reveal that consensus-based uncertainty estimators provide highly reliable indications of model confidence, while, in contrast, single-model methods such as temperature scaling show poor calibration for complex reasoning tasks. These methodology guidelines directly support our framework second level’s confirmatory inference approach where we leverage non-

parametric methods for testing demographic effects when LLM output distributions violate standard statistical assumptions.

These findings permit to ascertain crucial design implications: paired and token-matched estimates with explicit replication protocols and non-parametric statistical methods are essential for reliable LLM benchmarking. The field must direct itself towards comprehensive stability assessment which acknowledges fundamental uncertainties in model evaluation and reports confidence intervals rather than point estimates for all performance metrics.

2.2.6 Consensus Methods Under Token Budgets

Given the extensive instability sources identified in the previous section, consensus methods offer potential solutions from aggregation across multiple evaluation instances, although they introduce new challenges around budget constraints and bias propagation. Resource-constrained deployment environments necessitate optimization of consensus strategies, balancing performance gains against computational costs. Recent research has developed approaches for selecting and composing consensus, accounting for token budgets while maintaining fairness and quantifying uncertainty in consensus benefits.

Addressing budget constraints emerged as a main criterion in practical consensus methods for LLMs. Wang et al. [59] introduce their prompts to proxies framework, constructing compact LLM consensus with prompts selected to reproduce human population survey responses. Their method aggregates outputs from a small set of agents using majority voting and regression weighting. Wang et al. [59] demonstrate compact consensus can perform preference distributions approximation similar to real human survey responses studied over populations in both limited and unlimited budget settings. This raises a fundamental question of whether, if compact consensus can replicate these studies results, we are evaluating models' reasoning capabilities or their ability to reproduce statistical patterns from training data. Their results validate efficient consensus architectures for scalable alignment, which optimize both accuracy and outputs diversity without the need for expensive groups of models or large computation resources. Leiva-Araos et al. [60] further extend consensus methods to open-ended summarization problems and empirically report trade-offs between obtained accuracy, response diversity and computational cost. These results permit to underline the practicalities of budget-aware consensus design in maintaining robust performance while deploying LLM in real-world scenarios. However, their findings also contradict consensus theory, in which we expect complex aggregation systems to outperform simple averaging and suggest either a poor design in current "complex" methods or LLM outputs lacking the needed diversity to observe benefits from these techniques.

Further consensus optimization strategies leverage process reward alignment for LLMs. Park et al. [61] introduce their LE-MCTS framework using Monte Carlo tree search to coordinate models consensus. They compare token-level averaging, diversity maximization and other consensus selection strategies, along with persona instructions variations

under fixed budget constraints. After isolating each studied item and evaluating which consensus correctly answers on downstream tasks, they show that consensus selected for process diversity can outperform this best consensus selection and reduce token consumption by 40%. This effect is intensified when consensus are tuned to maximize reasoning patterns diversity. Wen et al. [62] introduce the BudgetThinker framework, which develops methods to select consensus of K prompts or profiles under strict token limitations and optimize reasoning accuracy and report the detailed selection. This work demonstrates dynamic consensus focused on query complexity can increase efficiency by 35% compared to consensus with fixed size. Wang et al. [5] propose evaluation under cost budget which selects consensus on Pareto frontiers optimizing cost, token usage and accuracy. This bootstrap validation approach aligns with the third level’s permutation testing for confirming consensus recommendations suggested from results in our framework. This is justified by the essentiality of statistical robustness for discerning performance gains from selection noise artifacts. Direct investigation by Han et al. [4] benchmarks performance compared against token limitations and show consensus composition has an impact on accuracy under several resource budgets scenarios, finally establishing that heterogeneous consensus in reasoning patterns can often outperform homogeneous ones.

Huang et al. [63] introduce the DeePEn framework, performing rigorous consensus design by combining probability distributions from heterogeneous LLMs at decoding stage, using aggregation by search. The method leverages bootstrap-based confidence estimation to provide robust uncertainty quantification, demonstrating that consensus performance benefits may be substantially overestimated if uncertainty is ignored. Their results show that carefully constructed heterogeneous consensus achieve strong, often Pareto-optimal tradeoffs between accuracy, robustness and calibration, even when consensus members have differing underlying architectures.

These developments collectively support the design implication that consensus selection on Pareto frontiers, with bootstrap or permutation corroboration and detailed documentation of consensus size and composition, represents best practice for budget-constrained deployment. These reporting requirements are integrated across all our framework levels: Level 1’s descriptive metrics, Level 2’s inferential robustness, Level 3’s consensus validation and Level 4’s stability assessment, as each level must address uncertainty quantification to ensure evaluation credibility.

2.2.7 Research Gaps

The synthesis of these research directions reveals significant progress while identifying current challenges in LLM benchmarking for normative reasoning. The systematic evaluation of test-time in-context learning regimes, demographic bias risks and consensus strategies under real-world resource budgets intensified conflicts between efficiency, fairness and reproducibility in model evaluation.

Empirical findings demonstrate protocol designs have large influence on accuracy, bias and reproducibility in normative datasets. The selec-

tion of allocated budgets and designed prompt templates introduce substantial variability into empirical outcomes, often exceeding 30%. Role-playing based evaluations jointly with cross-validation and uncertainty quantification identify performance disparities, systematically disadvantaging specific demographic groups. These findings highlight the risks of overfitting to particular populations in testing dataset, and how benchmarks can potentially encode rather than expose biases.

Measurable improvements can be observed in consensus approaches, where majority vote and selective consensus demonstrate from 20% to over 40% accuracy gains, along with meaningful decrease in demographic error rates, depending on task complexity and consensus composition. However, recent research such as Wang et al.’s work raise questions on whether compact consensus are evaluating reasoning capabilities or reproducing statistical patterns learned from training data. Additionally, contradiction between expected consensus decision theory and empirical findings suggest either a poor design of current complex techniques or insufficient reasoning diversity in LLM outputs to observe advanced consensus benefits.

Critical gaps remain despite these improvements in the field. Generalization of findings across benchmarks and datasets remains largely unestablished, accentuated by contamination detection inconsistencies reported by Fu et al. [29] with potentially drifting results as benchmarks evolve or reasoning structures emerge in models. The research field lacks standardized protocols for assessing whether improvements on a given benchmark generalize to related tasks or whether they represent overfitting to specific evaluation artifacts. The design validity remains weak for normative tasks and data, recalling the Theory of Mind and manipulation detection examples. It raises fundamental questions about how generalizable fairness results can be beyond controlled simulations. A clear separation between benchmark performance and real-world capabilities continues to expand as models become increasingly sophisticated for pattern-matching evaluation formats although lacking reasoning capabilities. As Riemer et al. [26] demonstrated for Theory of Mind evaluated tasks, models may outperform at benchmark pattern recognition but fail to display a flexible reasoning these benchmarks expect to measure.

Reproducibility remains sensitive to minor implementation details, such as label noise patterns, random seed selection, sample splitting procedures and library versions, all substantially affecting results. Current evaluation practices show difficulties to identify these variability sources as they interact in complex ways, reflecting a necessity for new protocols which would explicitly quantify and report several sources of uncertainty.

Evolution of research in this field requires addressing the methodology gaps identified across dataset quality and evolution stability. Designing standard approaches to quantify and identify uncertainty sources constitutes a promising direction. When applied systematically, nonparametric statistical methods offer solutions for managing distributional violations, often occurring in LLM outputs [58]. As label noise accounts for 30-40% of performance variance [55], standardized protocols necessity is further justified for handling disagreement between annotators and quantifying label uncertainty. However, no consensus framework exists for synthesiz-

ing these considerations into evaluation design.

A possible direction for improvements in the field would be to leverage consumption-aware consensus selection, combined with bootstrap validation and uncertainty quantification, for more robust and equitable LLMs assessment. However, theoretical advances must complete these methods, to increase our understanding of the limits of benchmark-based evaluation and the needed conditions to shift benchmark performances to real-world applications.

The implementation of these solutions introduces resource consumption that may limit their conformity to budget-constrained settings; therefore, it is necessary consider a trade-off between methodical evaluation and practical feasibility. As datasets, models and expectations continue to evolve with an increasing speed, ongoing methodological innovation and pluri-disciplinary discussions will become essential for maintaining a practical impact in deploying LLMs for reasoning in complex and high-stakes domain, while ensuring scientific integrity. This field lies at the junction where design choices about evaluation methodology will control our understanding of models capabilities jointly with fairness and reliability of AI systems deployed at scale. These findings motivate the design of both our benchmark and four-level evaluation framework, where descriptive assessment (Level 1), confirmatory inference (Level 2), consensus validation (Level 3) and stability exploration (Level 4) permits us to address gaps in methodology identified across dataset quality, bias assessment and reproducibility challenges.

Chapter 3

Benchmarking with Budget Constraints

3.1 Methodology

3.1.1 Overview and Research Questions

Building onto related work summarized in Chapter 2, since many benchmarks evaluate under unbounded token consumptions, this study investigates a performance comparison of different in-context learning strategies under realistic resources, addressing a gap in current benchmarking practices. We must define four research questions for this study: Firstly, which in-context learning strategy achieves the highest accuracy when constrained to a limited token budget? Secondly, can we identify a specific strategy recommendation based on the token-aware accuracy for each dataset-model pair? Finally, how do models rank against each other when evaluated under identical experiment settings?

3.1.2 Experimental Design

We conduct a controlled comparison across three dimensions: models, datasets, and strategies. This permits to isolate each component’s effect while maintaining our objectives (evaluate under resource constraints).

Dataset Selection

We evaluate on four datasets, selected to observe results on both normative and logical reasoning. The **MGSD** dataset [64] tests models’ ability to identify and report stereotypes presence in sentences. The **Mental-Manip** dataset . For each dataset, we selected a subset constituted of 500 binary classification samples with balanced labels, discerning *stereotype* from *unrelated* sentences for MGSD and *manipulative* from *non-manipulative* behaviors for MentalManip.

We also include two configurations of the **MMLU** dataset, which is a multi-label benchmark. Our standard MMLU subset contains 750 items balancing normative and non-normative reasoning: 50 samples from each of 10 normative subjects (comprising Professional Ethics and Moral Scenarios) and 5 logical reasoning subjects (comprising formal logic and

mathematics). This permits us to investigate if a given strategy’s performance on normative reasoning categories differs from logical categories. Our MMLU-Large configuration focuses on four subjects: 180 samples each from *Professional Law* and *Moral Scenarios*, and the complete *Formal Logic* and *College Mathematics* subjects, resulting in a total of 579 samples. It will allow us to compare and assess performances later in Chapter 4 with statistical significance.

Model Selection

Throughout this study, we benchmark six models selected to represent several LLM providers, architectures, approaches in training, advanced reasoning capabilities and region provenance. **GPT-4o-mini** and **GPT-4.1-mini** from OpenAI represent cost-effective models with advanced reasoning capabilities and large context windows [65, 66]. **Claude 3.7 Sonnet** from Anthropic has undergone training with Constitutional AI, an approach where the model is criticized based on principles such as the Declaration of Human Rights when trained [67]. Mistral’s **Small 3.1** and **3.2** provide European models with multilingual and multi-modal capabilities [68, 69]. The **DeepSeek-V3** model, designed in Asia, constitutes a large mixture-of-expert LLM architecture with strong reasoning capabilities [70]. This diversity permits to ensure our methods generalize beyond model’s specificities.

In-Context Learning Strategies

We standardize all in-context learning strategies across models and datasets, ensuring their fair comparison by defining the *CaseConfig* instance during implementation. We set the model temperature to 0.0 and the top- p parameter to 1.0 across each strategy for reproducibility. Table 3.1 provides a summary of the maximum allowed tokens and parameters for each strategy.

To avoid biasing the model’s reasoning, we only ask for simple labels such as *Yes*, *No* or capital letters (A, B, C or D) depending on the task. For instance, if the model is asked for a MGSD sample to return *unrelated*, there is no current method to be fully sure its definition of "unrelated" will not raise a conflict with its opinion on this question.

Zero-Shot This strategy provides our baseline. We set a limit of 300 tokens to the outputs. This strategy permits to test models’ ability to recall pre-trained knowledge without context.

Few-shot Learning We inject $2 \times K$ examples randomly sampled from the dataset (outside of the training set), *i.e.* K examples of each label to avoid biasing the model toward a single outcome. Similarly to zero-shot, the 300 token limit applies, although we must note the input prompt is longer due to the examples.

Chain-of-Thought Reasoning For this strategy, we increase the token limit to 700 to adapt to reasoning traces. We guide the LLM through the reasoning to obtain better answers (see Appendix).

Tree-of-Thought We implement a tree with fixed depth (2) and branching (3). The LLM attributes a score to each thought while generating permitting to extract the best path based on score. The final label is extracted from the highest score on this path if greater than the leaf’s score. We set a budget of 500 tokens for the generation process as final label selection does not require LLMs.

Generated Knowledge Prompting We attribute a budget of 300 tokens to generation and 200 tokens to label selection.

Definition Injection in Prompt As models struggle in defining clearly the concepts behind normative reasoning tasks, we also evaluate the Zero-shot and Few-shot strategy while injecting a designed definition to guide the LLM on the task. This is implemented for both MGSD and MentalManip datasets, but not for MMLU as it comprises too many broad subjects. For instance, it seems counter-intuitive to design a definition for Professional Law applicable to all presented samples without confusing the model. On the other hand, it is easy to define what represents a stereotype as the meaning can be easily visualized in all cultures. This permits to assess whether the obtained improvements stem from disambiguating the task by providing a small definition rather than from provided examples and prompt’s contents, eventually improving reproducibility under a fixed resource budget. These strategies will be named ZS-Def and FS-Def, "Def" standing for definition.

Tree-of-Thought with LLM-as-a-Judge The Tree-of-Thought strategy exhibits a few challenges: the generating model could grade its outputs implicitly only and reward its stylistic writing and position effects rather than its quality of reasoning. To assess the best reasoning path, we implement an alternative method ToT-Judge where the ToT is used only for generating the tree, then all paths are transmitted to a LLM-as-a-Judge deciding the correct path based on internal reasoning coherence and how this reasoning supports the final label. This approach separates generation from selection steps, resulting in an improved reproducibility under the resource budget.

Results Measurements

Our benchmark captures several signals to provide a descriptive overview of strategy-model pair’s performances under budget constraints. Evaluated metrics include: *accuracy*, *precision*, *recall*, and *F1*, all defined in Chapter 2. We also include standard deviation of the accuracy across strategies and track token costs.

3.2 Results

This section summarizes the observed results across the benchmark. Table 3.2 serves as a reminder of the performed experiments.

Strategy	Temp	Top- p	Max Tokens
Zero-shot (ZS)	0.0	1.0	300
Few-shot (FS)	0.0	1.0	300
Zero-shot Def (ZS-Def)	0.0	1.0	300
Few-shot Def (ZS-Def)	0.0	1.0	300
Chain-of-Thought (CoT)	0.0	1.0	700
Tree-of-Thought (ToT)	0.0	1.0	500
LLM-as-a-judge (for ToT-Judge)	0.0	1.0	256
GKP - Generation	0.0	1.0	300
GKP - Label Selection	0.0	1.0	200

Table 3.1: Summary of Parameters by Strategy

Dataset	Models	Strategies	Items	Configurations
MGSD	6	8	500	48
MentalManip	6	8	500	48
MMLU	6	8	750	48
MMLU-Large	6	8	579	48

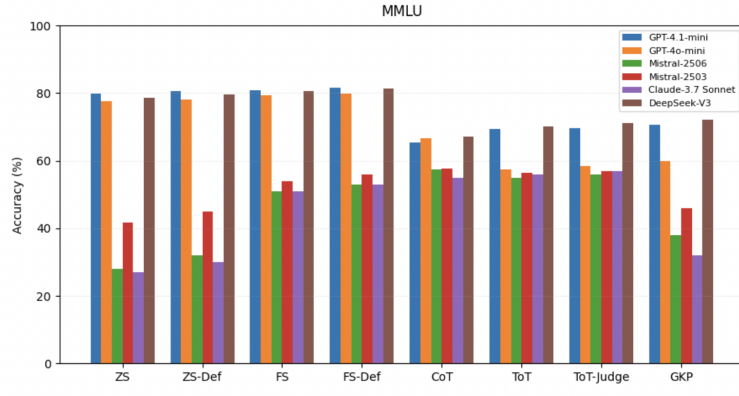
Table 3.2: Experimental Design Summary

3.2.1 Strategy Performance

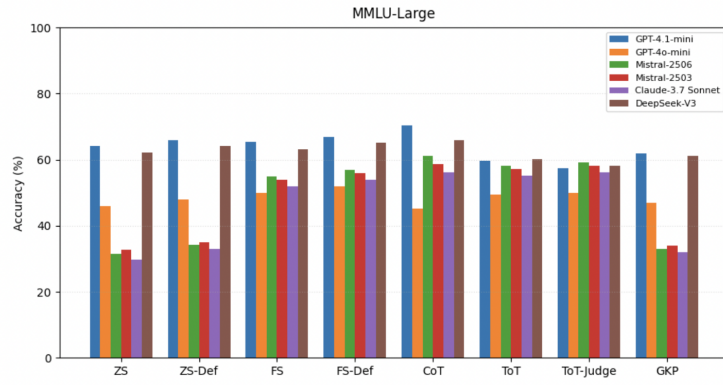
Few-shot prompting dominated the strategy rankings. FS-Def achieved the best mean rank (2.21) across model-dataset combinations, followed by FS (2.88). Zero-shot and ZS-Def strategies achieved high accuracies (ZS: 4.83, ZS-Def: 4.35) while consuming the least tokens. Multiple stage reasoning showed inconsistent gains in accuracy and higher token costs: CoT (4.69), ToT (5.71), and ToT-Judge (5.19). Generated Knowledge Prompting ranked last (6.15) and thus offered no significant advantage on both token consumption and accuracy.

Table 3.3: Strategy Rankings Across All Configurations

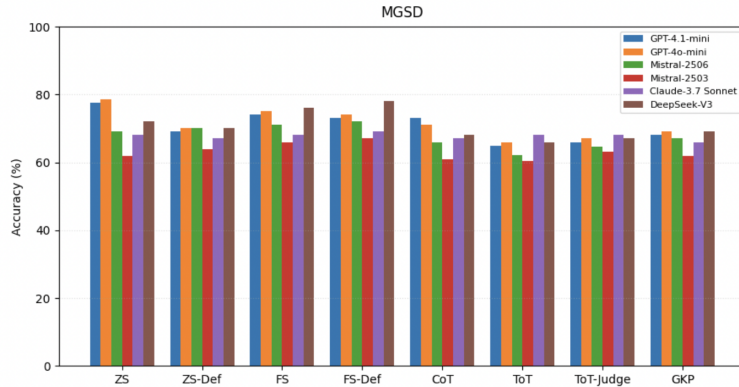
Strategy	Mean Rank	Best Dataset	Worst Dataset
FS-Def	2.21	MGSD	MMLU-Large
FS	2.88	MGSD	MMLU-Large
ZS-Def	4.35	MGSD	MMLU-Large
CoT	4.69	MGSD	MentalManip
ZS	4.83	MGSD	MMLU-Large
ToT-Judge	5.19	MGSD	MentalManip
ToT	5.71	MGSD	MentalManip
GKP	6.15	MGSD	MMLU-Large



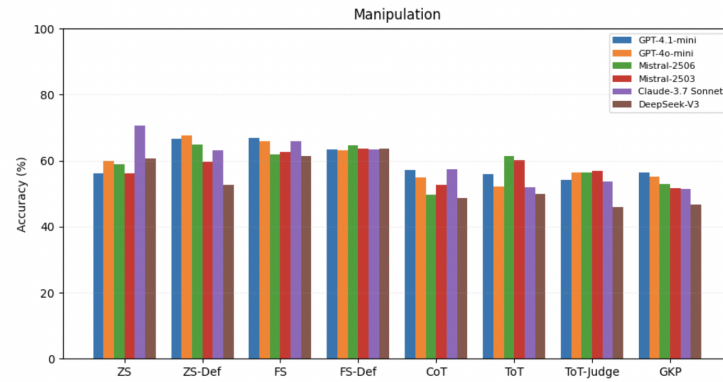
(a) MMLU



(b) MMLU-Large



(c) MGSD



(d) Manip

Figure 3.1: Overall Accuracy per Model-Strategy Pair (Across All Datasets)

3.2.2 Model Performance

GPT-4.1-mini (67.16%) outperformed other models, achieving the highest mean accuracy across all datasets along with the lowest variance in results ($\sigma=7.61$). Its performance was higher on MMLU and lower on MentalManip. GPT-4o-mini is ranking second (64.56%, $\sigma=8.70$). DeepSeek-V3 showed comparable performance (65.65%) but a higher sensitivity to studied dataset. The Mistral and Claude-3.7-Sonnet models scored lower ranks and higher variance across datasets, with accuracies ranging from 53.93% to 55.41%.

Table 3.4: Model Performance Summary (Percentage Scale)

Model	Mean Accuracy	$\sigma(\text{acc})$	Best Dataset	Worst Dataset
GPT-4.1-mini	67.16	7.61	MMLU	MentalManip
DeepSeek-V3	65.65	9.47	MMLU	MentalManip
GPT-4o-mini	64.56	8.70	MGSD	MMLU-Large
Mistral-Small-3.1	55.41	8.32	MGSD	MMLU-Large
Mistral-Small-3.2	55.17	11.91	MGSD	MMLU
Claude-3.7-Sonnet	53.93	13.08	MGSD	MMLU-Large

3.2.3 Dataset Patterns

MGSD and MentalManip

MGSD proved higher performances than MentalManip, where models achieved peak accuracies of 78.5% versus 70.73% respectively. Few-shot strategies consistently outperformed zero-shot across all models on MGSD evaluation. The opposite pattern was observed in MentalManip, as zero-shot often exceeded few-shot performance. This outcome suggests additional examples may introduce noise for manipulation detection tasks. Tree-of-Thought showed accuracy drops of 10 to 15% compared to simpler strategies on both binary classification tasks.

MMLU

MMLU showed the highest accuracies across all models and MMLU-Large the lowest. This performance gap suggests sensitivity to subject selection in dataset and possibly item selection. Few-shot methods still performed well on both MMLU datasets. CoT on the other hand showed mixed results which do not justify its token cost increase.

3.3 Discussion

3.3.1 Selection Guidelines

From our evaluation, we could identify three patterns. Firstly, few-shot prompting provides optimal trade-offs between accuracy and efficiency, ranking in the top two strategies across most datasets without token consumption of multiple-stage reasoning methods. Secondly, zero-shot

remains performing under strict token budgets as it achieves 90% of few-shot’s accuracy with 80% of its cost. Third, high reasoning strategies, such as CoT and ToT, show benefits on certain domains but lack results consistency compared to other strategies. GPT-4.1-mini appear as the optimal choice as it achieves the highest accuracy and lowest variance on this set of diverse tasks. Its performance consistency across all strategies suggests high-level internal capabilities. The accuracy gap on challenging datasets between GPT-4.1-mini and low-performance models (MMLU-Large, MentalManip) strengthen this high capability hypothesis.

Our results support the following approach for strategy selection:

- Zero-shot should be used as default for large-scale evaluation, with advices to upgrade to few-shot when accuracy exceeds cost thresholds;
- Deploying FS-Def to balance cost and efficiency offers the best mean performance with acceptable token costs.

Finally, the observation that definition injection (ZS-Def and FS-Def) often improved performance suggests that task ambiguity may be a limiting factor to model accuracy rather than reasoning capability. Study 2 will investigate if demographic profiles implicitly provide similar results and potentially explain any observed differences in performance.

3.3.2 Dataset Characteristics

Binary and multiclass datasets showed a higher contrast than contents. Binary tasks rewarded concise, example-based prompting, while multiclass problems showed mixed responses to reasoning elaboration. Differences in MGSD and MentalManip performances suggest a necessary dataset-specific prompt optimization. MentalManip’s low improvement on few-shot prompting may reflect ambiguity of the task which cannot be resolved through additional examples.

3.3.3 Limitations

Our sample selection for dataset design is non-aleatoric and fixed. This eliminates output variance from the LLMs but may not reflect real-world performance ranges. Temperature being set to 0 may also limit the ability for the models to present diversified reasoning as discussed in Chapter 2, resulting in a lower performance.

The benchmark may suffer from pre-training exposure as datasets are public. Task ambiguity may also be a limiting factor, as label noise (corresponding to annotators disagreement) can limit the highest achievable performance. Models development’s pace could make our benchmark inaccurate as it may not capture emerging capabilities in new models.

Chapter Summary

This chapter established a baseline performance benchmarks for six models across eight in-context learning strategies subject to a fixed token

budgets. It addresses a critical gap in budget-aware benchmarking and evaluation. Through comparison on four datasets, we identified three key findings necessary to inform next studies.

Dataset patterns revealed important contrasts. Few-shot learning performed well on MGSD (5-10% improvement over zero-shot), while showing lowered performance with additional examples on MentalManip. This suggests ambiguity in manipulation detection that cannot be solved by providing demonstration.

Multi-step reasoning strategies (Chain-of-Thought and Tree-of-Thought) showed no consistent accuracy gains, while Generated Knowledge Prompting ranked last overall offering no advantages. These negative results permit to establish when complex prompting strategies adds value.

Implications for Next Studies

These baseline results permit to inform the design of Studies 2 and 3 in the following ways. Regarding the model selection, GPT-4.1-mini’s performance and consistency justify its use for demographic analysis in Study 2. We focus only on the highest performing strategies, the definition-augmented FS. Reducing from 48 to 4 configurations (of 60 profiles) per dataset permits deeper investigation of profile effects while maintaining manageable study design.

We identified accuracy limits for each dataset (78.5% for MGSD, 70.7% for MentalManip, 81% for MMLU), providing reference performances for demographic role-playing comparison. The reported variance for each strategies provides descriptive expected noise levels. Study 2’s framework will implement different methods to discern demographic effects from random variation.

The observed variance patterns establishes the baseline instability, which will be studied in Study 3. Study 3’s cross-validation framework will ascertain whether demographic profiles exhibit consistent patterns or if they simply add noise to the existing variance. The finding that GPT-4.1-mini exhibits the lowest variance ($\sigma=7.61$) compared to other models may suggest that stability itself is a measurable model characteristic. Study 3 will investigate whether certain demographic profiles increase or decrease this stability and potentially identify low variance profiles.

Chapter 4

Profile Inference and Consensus Selection

This study introduces the first three levels of our statistical analysis framework to evaluate how demographic role-playing across gender, ethnicity and age affects model reliability under a single model and whether these differences vary with in-context learning strategies (zero-shot, few-shot, chain-of-thought). The process is divided in two parts, the first part corresponds to modeling the association between profile traits and per-sample correctness of answers using a Generalised Linear Model (GLM) in Levels 1 and 2. The second part corresponds to selecting majority-vote consensus over subsets of profiles in Level 3, comparing them to a baseline without role-playing from the previous study.

4.1 Methodology

4.1.1 Data, models and strategies

We narrow our study to *GPT-4.1 mini* model only, using the Few-shot strategy from Study 3. This strategy is again implemented onto MGSD, MentalManip and MMLU datasets, using the same sample sets and experimental settings used in Chapter 3.

We run the model with a system prompt stating no specified demographics (the same system prompt as in Chapter 3), and follow with execution using 60 profiles prompts, systematically varying on three traits detailed as following:

1. Ethnicity: White, Black, Asian, Indian, Middle Eastern, and Latine;
2. Gender: Men, Women;
3. Age: 20, 25, 35, 45, 55

We will always compare role-playing profiles (across demographic traits) compared to a neutral baseline p_0 . The analysis is performed on separate pairs of dataset and strategy.

4.1.2 Level 1: Descriptive Analysis

Level 1 describes with light assumptions the performance across different traits of profiles with uncertainty quantification and decomposes errors compared to the baseline profile to inform modeling and selection in Levels 2 and 3. Results justify the implementation of hypothesis testing with multiplicity control in Level 2.

Exploration of Contrasts between Traits

To ascertain meaningful gaps between trait groups independently of model structure, we implement tests to observe differences in distributions of results with item-clustered bootstrap. They permit to evaluate which factors need to be modeled in Level 2, and which differences in accuracy are too unstable to justify their modeling.

For each trait group G , we calculate its empirical group-level accuracy $\hat{a}_G$ (see Appendix B.2), corresponding to the mean of accuracy per profile within a given group for all profiles in the group G .

We use it to compare $\hat{a}_{G_1} - \hat{a}_{G_2}$ difference between two trait groups using a bootstrap process clustered by item. We implement the bootstrap process by sampling with replacement $B = 1000$ samples (using a fixed seed), recomputing the difference at each iteration and reporting the 2.5th and 97.5th percentile in the distribution of results. We use additionally the Cohen's h distance to report effect size independent of scale (see Appendix B.2).

The tails of the bootstrap distribution and multiple comparisons (such as FDR and Holm) are displayed as results, but only used later in the Level 2 for inference.

Error Directionality

We compare each given profile p against a fixed baseline p_0 using two rates we introduce: the *Error correction* and *Extra-Errors*. EC represents the baseline errors that p corrects and EE the baseline's correct predictions p turns into errors. They are defined in Chapter 2 by the paired outcomes $(Y_{i,p_0}, Y_{i,p})$. We report $EC(p|p_0)$ and $EE(p|p_0)$ alongside accuracy and net accuracy changes $\Delta acc(p; p_0)$:

$$\Delta acc(p; p_0) = \frac{1}{|\mathcal{J}|} \sum_{i \in \mathcal{J}} (Y_{i,p} - Y_{i,p_0}) \quad (4.1.1)$$

We then use McNemar's test to assess directional asymmetry of the EC and EE .

Throughout the study, we consider two main sources of variability in the results: the variation across items within a profile, where items are considered as binomial; and the variation of results between different profiles within a trait group.

EC , EE and the net accuracy change $\Delta acc(p; p_0)$ are all formally defined in Appendix B.2. We also define there McNemar's test, exact p and associated confidence intervals for the error directionality case.

High-disagreement evaluation

We define the disagreement score as:

$$D = 1 - \max_k \frac{n_k}{P} \quad n_k = \quad (4.1.2)$$

representing the fraction of profiles predicting a different label from the majority's prediction (see Appendix B.2).

We associate the D score to Shannon Entropy H of the empirical prediction distribution:

$$H = - \sum_{k \in \mathcal{K}} \frac{n_k}{P} \log_2 \left(\frac{n_k}{P} \right) \quad (4.1.3)$$

We report items with $D > \tau$ (*e.g.* $\tau = 0.7$) to assess qualitatively the confidence of the profiles on their prediction and the difficulty of the evaluated item.

4.1.3 Level 2: Confirmatory Inference

Level 2 assess whether the signals from Level 1 persist under modeling accommodating to data structure. We fit a binomial generalized linear model (GLM) with logit link to profile-level rows with traits as covariates. The model's equation is:

$$\Pr(Y_{ip} = 1) = \text{logit}^{-1} \{ \beta_0 + \beta^\top G_i + \theta^\top (G_i \otimes G_i) \},$$

where G_i represents a trait factors encoded as a dummy variable with pre-specified levels (*e.g.* men or women for gender). To avoid unstable estimates, we include pairwise interactions ($G_i \otimes G_i$) only if sufficient data is supporting.

Coefficients are estimated by maximum likelihood. As correlation and heteroskedasticity at sample-levels are artificially designed, we use cluster-robust covariance with clustering on item and a small-sample degrees-of-freedom correction.

For each trait's main effect and for each interaction block, we assess the null hypothesis using a Wald statistic (see Chapter 2). Within traits, we also test pairwise differences between levels against odds on log-scale but refer to them as odds ratio for simplicity.

We control multiple testing (of simultaneous null hypothesis) using the BenjaminiHochberg false discovery rate procedure, applied with families. This process consists in associating one family for all block tests (main effects and interactions) and separately fixing one family per trait for the set of pairwise contrasts within that trait. It preserves statistical power within coherent sets of hypotheses and maintains transparency on error rates. We include in Level 2's outputs coefficient table with cluster-robust SEs, z/t statistics, the block-test with degrees of freedom and p -values, odds ratios and CIs, FDR-adjusted pairwise comparisons, and designed plots of marginalized probabilities with 95% confidence intervals.

These results permit to infer confirmation on demographic role-playing effects in normative reasoning tasks.

4.1.4 Level 3: Nonparametric Verification

Level 3’s purpose is to turn statistical patterns into decisions about system recommendations based on these tests with minimal assumptions. To ensure robustness in the design, we separate the items into a two fold splits on stratified folds once again. The first fold is used only to select among candidate profiles and groups. The second fold is used for confirming these results by reporting performance.

Candidate profiles are consensus formed by grouping profiles on shared trait keys and aggregating their predictions by majority vote (from already computed results). Baseline serves as tie-breaker for these consensus. We compute overall accuracy, the improvement relative to the baseline Δ_{acc} and the error correction and extra-error rates defined in Level 1.

Selection for deployable consensus is performed by defining a Pareto frontier between minimized extra-error rate (aimed to be minimized) and accuracy (aimed to be maximized).

Uncertainty Quantification

We implement an item-paired bootstrap, resampling each sample with replacement while preserving dependence across profiles in each sample. We report two-sided bootstrap p -values for Δ_{acc} and 95% confidence intervals for accuracy, rescue rate, and extra-error rate.

The same bootstrap is applied within each category with sample size above a threshold, *i.e.* 50 samples, preventing over-interpretion of sparse data. Afterwise, BenjaminiHochberg FDR is controlled across consensus and categories for three parameters: overall improvement in accuracy, directional errors providing more error correction than extra-errors, and the opposite directionality in errors.

To verify our hypothesis on trait-level signals, we perform item permutations tests. For a two-group comparison, we use the mean difference of per-item correctness as main statistic. For more than two groups, we implement a F -statistic averaged across items. Both cases implement random permutations of trait labels within items, after which the statistic is recomputed forming a permutation p -value. FDR is again controlled across the families of traits involved in this verification step. Procedures for Multiplicity are applied within families at each Level as described above, for well-defined families.

We discern confirmation and descriptive results in figure captions and tables.

4.2 Results

This study evaluated demographic role-playing effects on GPT-4.1-mini using few-shot prompting across 60 profiles (varying by gender, ethnicity, and age) on the MGSD, MentalManip, MMLU and MMLU-Large

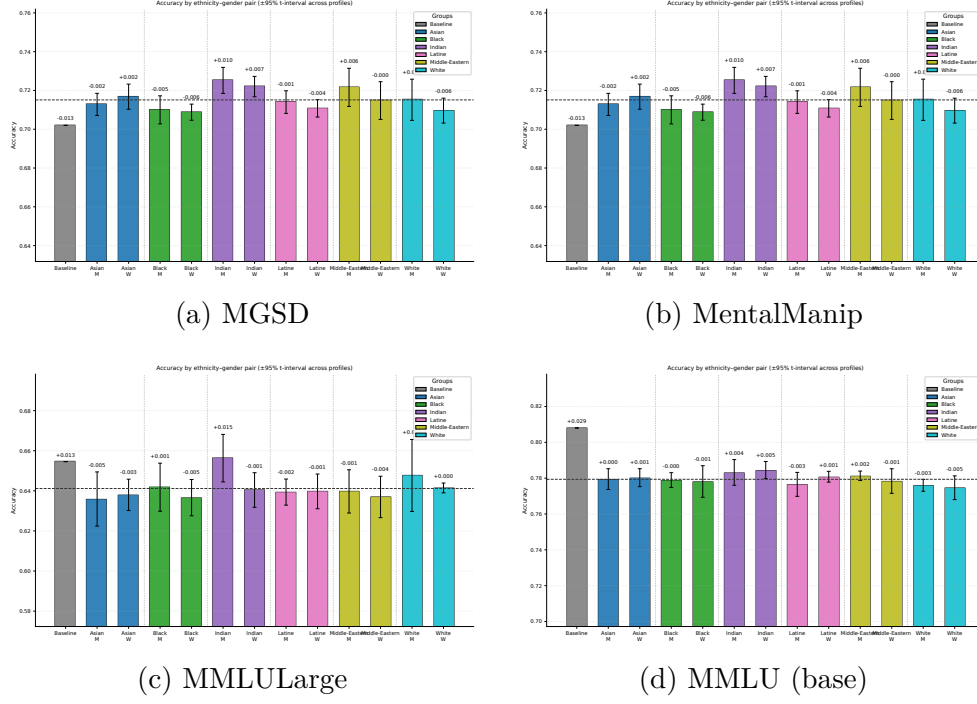


Figure 4.1: Accuracy by ethnicity-gender pair with 95% CIs across datasets. M: Men, W: Women

datasets. The analysis reveals subtle patterns on the demographic framing influence on model behavior on several tasks.

Level 1: Descriptive Analysis

Throughout Level 1 results, the datasets did not reveal conclusive results on the demographic disparities in results.

Effect sizes Regarding the measured effect sizes in MGSD dataset results, we observe several ethnicity gaps around 1 to 3%. The non-zero pairwise gaps show Indian profiles perform better than other ethnicities, with the highest gap being 2.8% between Black profiles and Indian profiles. Age has no effects on the results and reveals itself to be negligible. However, all p -values do not exceed the 0.05 threshold for significance. All the obtained effect sizes are summarized in Table MentalManip and MMLU dataset do not get more significant results as all Cohen’s h distances are below the 0.2 threshold to assert some significance, but we can still observe the difference Black and Indian performance with 1.4% in accuracy difference. We once again observe in each dataset the difference gap between Black and Indian ethnicities to be included in the 5 highest gaps.

Figure 4.1 show some non-uniform results in the performances. Other metrics in preliminary analysis do not permit us to draw conclusive interpretations on the presence of demographic disparities question.

However, the subtle but systematic disparities in the effect sizes (stigmatizing Black profiles in lower results than Indian) motivate us to study deeper the effects of demographics on the results.

Level 2: Confirmatory Inference

GLM results While MentalManip, MMLU and MMLU-Large displayed no significant effects during GLM modeling, Level 2’s GLM blocks resulted in a more nuanced picture for MGSD dataset. The GLM analysis showed ethnicity displayed some marginal effects, with a F -statistic equal to $F(5, 499) = 2.13$ and a p -value $p \approx 0.061$ closer to the 0.05 threshold, closely followed by gender ($p \approx 0.10$) also approaching it. Interactions effects between multiple trait groups were statistically non-significant ($p \approx 0.36$). The main observed results from modelisation are summarized in Tables 4.1 to 4.4.

Table 4.1: Level 2: GLM Wald tests clustered on samples for MGSD. Predicted outcome: Correctness.

Block	F	df1	df2	p
main: C(ethnicity)	2.127	5	499	0.0609
main: C(gender)	2.681	1	499	0.1022
inter: C(gender):C(ethnicity)	1.100	5	499	0.3595

Table 4.2: Level 2: GLM Wald tests clustered on samples for MentalManip. Predicted outcome: Correctness.

Block	F	df1	df2	p
main: C(gender)	1.057	1	419	0.3044
main: C(ethnicity)	0.639	5	419	0.6703
inter: C(gender):C(ethnicity)	0.442	5	419	0.8193

Table 4.3: Level 2: GLM Wald tests clustered on samples for MMLU. Predicted outcome: Correctness.

Block	F	df1	df2	p
inter: C(gender):C(age)	1.451	4	749	0.2156
inter: C(gender):C(ethnicity)	0.778	5	749	0.5659
main: C(age)	0.632	4	749	0.6398
inter: C(ethnicity):C(age)	0.800	20	749	0.7148
main: C(gender)	0.132	1	749	0.7165
main: C(ethnicity)	0.492	5	749	0.7821

Table 4.4: Level 2: GLM Wald tests clustered on samples for MMLU-Large. Predicted outcome: Correctness.

Block	F	df1	df2	p
main: C(ethnicity)	1.361	5	578	0.2373
inter: C(gender):C(ethnicity)	0.927	5	578	0.4630
main: C(gender)	0.328	1	578	0.5671

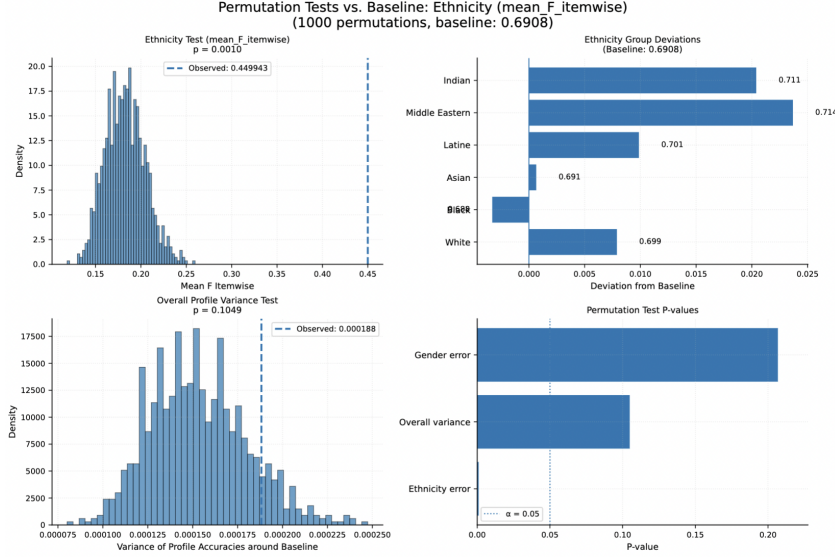


Figure 4.2: Permutation Tests on Ethnicity for MGSD dataset

Level 3: Nonparametric Verification

Permutation Test results Our results using permutation tests resulted in a highly significant result ($p = 0.001$) for MGSD dataset. This result directly contrasts with GLM’s findings which were non-significant. This observation uncovers methodological insights: parametric tests can miss non-linear effects which can be revealed by non-parametric tests like permutation test. This result suggests demographic role-playing creates non-random deviations with low magnitudes but do not impact on the overall performance in a significant manner.

The descriptive statistics performed in the permutation test confirm these patterns. The largest absolute accuracy difference in MGSD was measured between Indian (72.0%) and Black profiles (69.2%). In contrast, the gender differences were low, with highest difference of 0.5% between Men (71.0%) and Women (70.4%) performances. Although these differences are small, the permutation test raises a need for deeper interpretation due to their statistical significance.

Similarly, for MMLU-Large, we also observe some significant permutation tests results. The observed p -value measured was 0.001 for permutation on ethnicity. This suggests very significant results, with around 4 to 6% of accuracy variations below the baseline’s performance. This suggests some non-beneficial effect of ethnicity on MMLU-Large. The permutation tests plots in Figures ?? to ?? show the permutations tests for MMLU-Large and MGSD on gender and ethnicity. We can see the overall distribution of results

4.3 Discussion

Across MGSD, MentalManip, MMLU, and MMLU-Large, our results consistently demonstrate small and inconsistent performance gaps across gender and ethnicity dimensions. GLM block tests yield non-significant results on all datasets, while descriptive pairwise gaps remain of low magnitude throughout. When effects are detected, such as in the MGSD

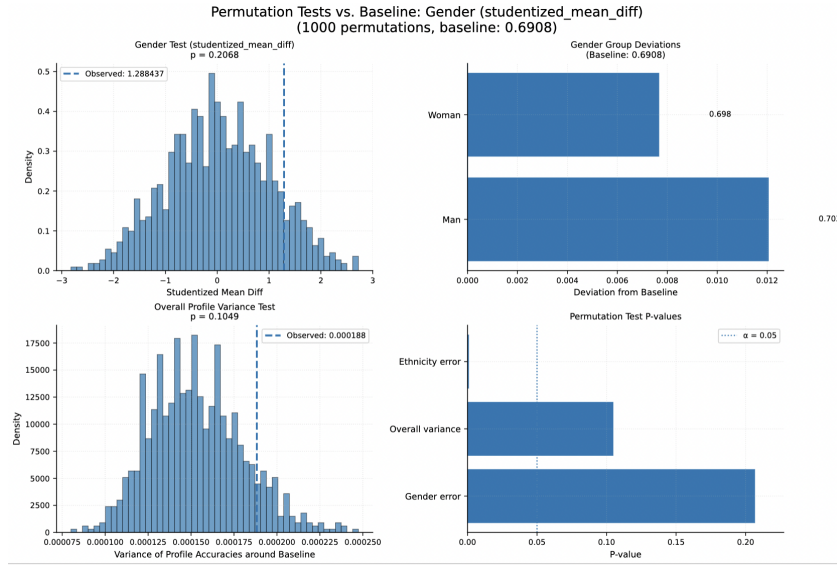


Figure 4.3: Permutation Tests on Gender for MGSD dataset

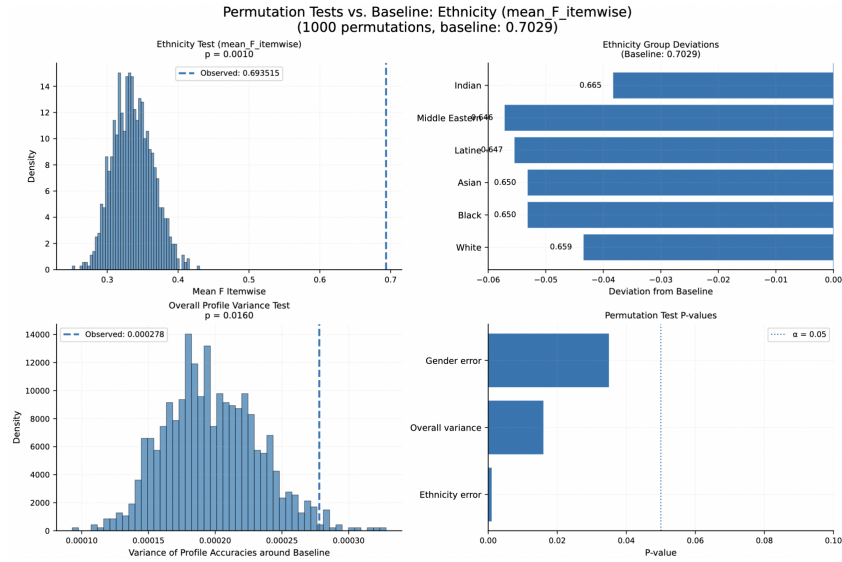


Figure 4.4: Permutation Tests on Ethnicity for MMLU-Large dataset

ethnicity permutation test, the effect sizes remain small and fail to reflect consistent performance differences across demographic groups.

The distinct pattern observed in MGSD merits particular attention and can be attributed to several factors. Most notably, the model exhibits substantial performance variation across different stereotype categories, with gender stereotypes achieving only 63% accuracy compared to 75% for profession stereotypes. This 12% difference dwarfs the maximum 2.8% gap observed between ethnic groups, suggesting that the subjective nature of content poses a far greater challenge than demographic framing. This finding aligns with the limitations of MGSD discussed in Chapter 2, where lack of context emerges as the primary obstacle in normative reasoning benchmarks. The model’s struggle with ambiguous concepts like gender stereotypes, compared to less contested categories such as professional stereotypes, indicates that demographic framing cannot address these fundamental task limitations. Rather than introducing systematic

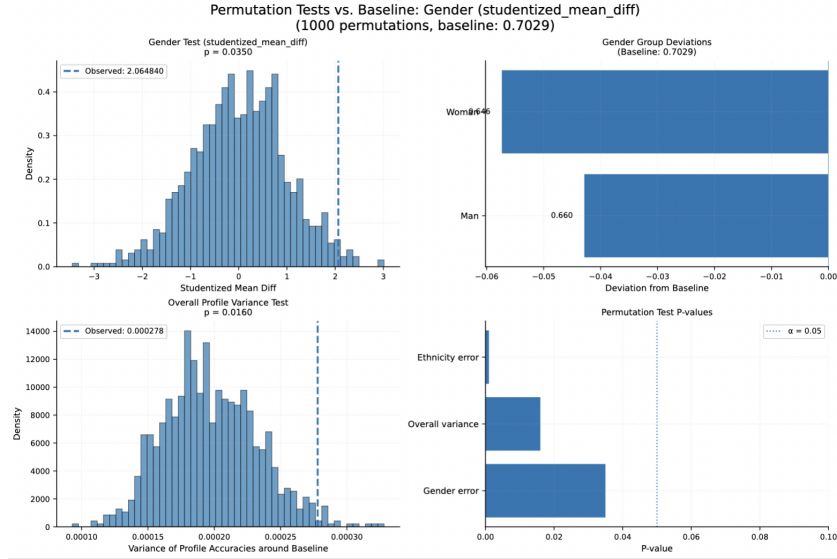


Figure 4.5: Permutation Tests on Gender for MMLU-Large dataset

bias, the model appears to accurately reflect the inherent noise present in human-generated training data and example patterns.

The systematic performance advantage observed for Indian and Middle Eastern profiles warrants careful interpretation. One possibility involves linguistic nuance, where the specific structure of test samples may align more closely with the model’s learned representations of these ethnicities. Additionally, patterns in the pre-training data may have created associations between certain demographic indicators and reasoning approaches. The system prompt defining these profiles could therefore trigger distinct internal reasoning pathways, potentially leading to more cautious or robust reasoning strategies for specific demographics. However, these effects remain subtle and operationally minor.

These findings carry important implications for how we conceptualize and evaluate fairness in large language models. Defining fairness solely through the absence of performance disparities on objective tasks proves insufficient when models must handle ambiguous, subjective content. Our results demonstrate that a model can satisfy traditional fairness metrics while still exhibiting subtle, non-random patterns detectable only through sensitive statistical tests. This suggests we need more nuanced definitions of fairness that incorporate behavioral consistency alongside performance equity.

Furthermore, our analysis strongly indicates that the primary challenge lies not in the model’s differential response to demographic prompts, but in the inherent limitations and ambiguities of the evaluation datasets themselves. Rather than engineering increasingly complex solutions to address minor demographic effects, research efforts would be better directed toward developing more precise benchmarks with less ambiguous ground truth labels. Such improvements would enhance model performance across all user groups rather than attempting to compensate for dataset limitations through demographic adjustments.

The discrepancy between GLM and permutation test results, combined with our findings on consistency and boldness relationships from

Study 3, argues for a fundamental shift in evaluation methodology. Simple accuracy scores fail to capture the complex behavioral patterns that emerge under different prompting conditions. Instead, we need multi-tiered evaluation frameworks that incorporate operational tests and behavioral metrics capable of detecting subtle, non-linear effects. While these effects may not substantially impact overall accuracy, they could become significant in high-stakes applications where even small systematic biases carry serious consequences. This more sophisticated approach to evaluation will be essential as we deploy language models in increasingly sensitive and consequential domains.

Chapter Summary

This study provides strong evidence that GPT-4.1-mini’s reasoning is primarily driven by query content rather than demographic context across both objective and normative tasks. Our comprehensive evaluation reveals that when using few-shot prompting, the model maintains consistent performance across diverse demographic profiles, with task characteristics and content difficulty emerging as the dominant factors determining outcomes.

While permutation tests on the MGSD dataset detected subtle, statistically significant patterns, these effects remained consistently small and operationally negligible. The magnitude of performance variation attributable to demographic framing pales in comparison to the variance explained by category-specific challenges and inherent task ambiguity. For instance, the model’s differential performance across stereotype categories far exceeds any gaps observed between demographic groups, suggesting that addressing fundamental task limitations would yield greater improvements than demographic-based interventions.

These findings indicate that the model exhibits an inherent fairness that should be preserved rather than disrupted through unnecessary demographic adjustments. Rather than engineering solutions for minor demographic effects, research efforts should prioritize reducing task ambiguity and improving benchmark quality, as these factors account for substantially more performance variance. The results establish a strong empirical foundation for shifting discussions of AI fairness from broad generalizations toward nuanced, task-specific, and data-driven analysis that recognizes the primacy of content difficulty over demographic framing in determining model behavior.

Chapter 5

Stability of Profile Effects

This study follows the previous study by focusing on the central problem to understand how consistency in results and "boldness" vary across profiles. We evaluate whether bolder behavior exhibit better predictions without introducing new mistaken predictions. We also assess the extent to which the effects can be associated to demographic traits and compared against other profile traits. We define boldness as a profile's predictions going against majority prediction across consensus. We evaluate variability across stratified cross-validation folds. Finally, we build an association model to assess the statistical links between demographic traits and performance metrics (*e.g.* accuracy), boldness, benefit-risk trade-offs.

5.1 Methodology

5.1.1 Data structure, baseline, and outcomes

We use the same experiment settings as previous study in terms of model, strategies and datasets: GPT-4.1-mini on MGSD, MMLU, MMLU-Large and MentalManip.

We set fixed seeds for randomized tests. Multiplicity is controlled within predeclared families via Benjamini-Hochberg FDR at 5%.

5.1.2 Ambiguity and boldness metrics

Boldness is a concept we will try in this section to define for a given consensus of LLMs. In this section, we define three new metrics for boldness attributed to a single profile: **COI**, **ATI** and **CAI**. We compute a **leave-one-out** consensus for each item to define and measure boldness, through aggregation of predictions across all profiles except the profile being scored. Appendix For the evaluation of profile p , we denote the consensus label by $\hat{C}_i^{(-p)}$ and its strength by $S_i^{(-p)} \in [0.5; 1]$ defined as the proportion of votes sharing the most selected label when excluding p 's prediction. We also define ambiguity as $A_i^{-p} = 1 - S_i^{(-p)}$.

These metrics lay a basis for implementing the **Consensus-Oriented Intervention** defined as the weighted probability that a profile p disagrees with the baseline on samples where the consensus supports the

baseline's prediction. This is formulated as:

$$\text{COI}_p = \frac{\sum_{i \in \mathcal{J}} S_i^{(-p)} F_i^{(p)} M_i^{(-p)}}{\sum_{i \in \mathcal{J}} S_i^{(-p)} M_i^{(-p)}} \quad (5.1.1)$$

where $M_i^{(-p)}$ and $F_i^{(p)}$ are indicator functions for *consensus agreement with baseline* and *profile disagreement with baseline* respectively.

We define ambiguity as $A_i^{-p} = 1 - S_i^{(-p)}$.

Two secondary metrics implemented is the Ambiguity Targeting Index (ATI), which assesses the location of boldness. We define it as the difference of profiles disagreement rate between the highest and lowest quartile of ambiguity $A_i^{(-p)}$ (p excluded). Let $Q_{0.25}$ and $Q_{0.75}$ be the 25th and 75th percentiles respectively. Then the corresponding quartiles are the sets of samples:

$$\mathcal{J}_{0.25} = \{i : A_i^{-p} \leq Q_{0.25}\}$$

and:

$$\mathcal{J}_{0.75} = \{i : A_i^{-p} \geq Q_{0.75}\}$$

We define then the ATI as:

$$\text{ATI}_p = \frac{1}{|\mathcal{J}_{0.75}|} \sum_{i \in \mathcal{J}_{0.75}} F_i^{(p)} - \frac{1}{|\mathcal{J}_{0.25}|} \sum_{i \in \mathcal{J}_{0.25}} F_i^{(p)} \quad (5.1.2)$$

Finally, the Consensus-weighted Ambiguity Index (CAI) is the expected disagreement weighted by ambiguity.

$$\text{CAI}_p = \frac{\sum_{i \in \mathcal{J}} A_i^{(-p)} F_i^{(p)}}{\sum_{i \in \mathcal{J}} A_i^{(-p)}} \quad (5.1.3)$$

5.1.3 Consistency across Folds

Another concept we will focus on in this study is the consistency of models. Result's stability is assessed through a stratified K-fold cross-validation on samples, which allows for reliable tests by mirroring the dataset's label distribution. We compute the fold's accuracy for each profile and define volatility $\sigma_{\text{fold}(\text{acc})}$ as the standard deviation of accuracy across folds, which we already implemented in Chapter 3. Consistency is then defined as:

$$\text{Consistency} = \frac{1}{1 + \sigma_{\text{fold}(\text{acc})}} \quad (5.1.4)$$

Throughout the folds, we track the error correction (or rescue) EC and extra-error EE rates to assess if the stability has a positive impact. We report both the means and pooled estimators.

5.1.4 Statistical Analysis

Correlation

We estimate Pearson correlations between the following: volatility, boldness metrics (COI, ATI and CAI), accuracy, error correction (rescue), and extra-error at profile level. We compute two-sided p-values for sample size greater than 3 and control FDR across the correlation family.

Profile Classification

To ascertain possible effects associated with demographic traits, profiles are divided by splits on median on the basis of the volatility and COI metrics into two dimensions: Consistent - Bold, Inconsistent - Bold, Consistent - Cautious, Inconsistent - Cautious.

Nonparametric Tests

We primarily compare highly-volatile profiles' extra-error and error correction (rescue) rates within demographic group. We also report Cohens h with bootstrap CIs. We ascertain significance using a Monte-Carlo permutation test. Welch's t -test and Cohens d are reported to serve as sensitivity metrics. Tests on subgroups per trait are FDR-controlled.

5.2 Predictive Attribution Modeling

We now define methods to quantify the amount of variation in boldness, benefitrisk, performance, samples disagreement, and efficiency which can be attributed to (thus explained by) demographic traits compared to other profile traits.

Variance Decomposition

For each outcome, we fit an ordinary least squares model (OLS) on a design matrix of trait dummies at profile level. The outcome we fit include the accuracy, error correction, extra-error, COI, ATI, CAI and disagreement rate.

We then use Shapley/LMG two player decomposition of variance to obtain each blocks unique contribution: both demographics alone and demographics added last, and reproducing the procedure for the other traits blocks. This permits to assess the contribution of demographic traits on the result's variance.

Uncertainty quantification.

We quantify uncertainty by implementing a bootstrap at profile level for unique R^2 contributions, resampling profiles and recomputing $R^2_{\text{demo, unique}}$, $R^2_{\text{other, unique}}$, R^2_{full} and D to form percentile CIs. We report the raw and standardized coefficients such as R^2 from K -fold cross-validation on full model, with standardization by the ratio of predictor and outcome standard deviations. We flag outcomes where the dominance ratio CI exceeds 1.

5.2.1 Normative assessment of boldness via non-parametric tests

We finally assess whether the volatility of profiles delivers higher error corrections than low volatility profiles within the same demographic traits. This is performed by measuring Cohen's h on error correction

rates with non-parametric uncertainty as the significance is evaluated by a permutation test.

To provide a robust sensitivity-aware evaluation, we also provide Welch's unequal-variance t -test and Cohens d distance with bootstrap confidence intervals.

5.3 Results

Consistency-Boldness Analysis

The relationship between profile consistency and boldness metrics revealed complex, dataset-dependent patterns that challenge simple interpretations. MMLU and MGSD exhibited significant negative correlations between volatility and boldness ($r = -0.316, p = 0.014$ and $r = -0.301, p = 0.019$ respectively), confirming that consistent profiles tend to make predictions deviating from consensus in these domains. This pattern validates our theoretical framework linking stability to decision confidence. However, MentalManip and MMLU-Large showed negligible correlations ($r = -0.063, p = 0.632$ and $r = -0.050, p = 0.705$), suggesting that our boldness conceptualization may be less applicable to certain task types where consensus itself proves unreliable.

The relationship between accuracy and boldness demonstrated even more pronounced task-specificity. MMLU showed a strong positive correlation ($r = 0.528, p < 0.001$), indicating that profiles willing to deviate from consensus achieved superior performance on academic knowledge tasks. This finding suggests that strategic non-conformity can improve outcomes when consensus reflects average rather than optimal reasoning. MGSD exhibited a weaker positive trend ($r = 0.217, p = 0.097$) that failed to reach significance, while MentalManip showed a significant negative correlation ($r = -0.282, p = 0.029$), warning that boldness in manipulation detection carries genuine risks. MMLU-Large demonstrated minimal association ($r = -0.161, p = 0.220$), further highlighting the dataset-dependent nature of these relationships.

Trait-Specific Patterns

Analysis of consistency-boldness correlations across demographic traits revealed striking heterogeneity that defies simple categorization. Among ethnic groups, White and Indian profiles on MMLU showed strong negative correlations ($r = -0.789, p = 0.007$ and $r = -0.761, p = 0.011$ respectively), while Middle Eastern profiles exhibited similar patterns on MGSD ($r = -0.808, p = 0.005$) and Asian profiles on MMLU-Large ($r = -0.674, p = 0.033$). These strong associations suggest that certain demographic framings may trigger more decisive reasoning patterns, though the mechanism remains unclear. Black profiles showed inconsistent patterns across datasets, while Latino profiles exhibited no discernible relationship, indicating that demographic effects on reasoning

consistency are neither uniform nor predictable.

Age demonstrated particularly intriguing patterns, with 45-year-old profiles showing significant negative correlations on both MMLU ($r = -0.597, p = 0.041$) and approaching significance on MGSD ($r = -0.516, p = 0.086$). This age-specific effect suggests that middle-age framing may invoke particular decision-making patterns, possibly reflecting assumptions about accumulated experience or cognitive maturity embedded in the model’s training data.

Archetype Distribution and Characteristics

Classification of profiles into behavioral archetypes based on median splits of consistency and boldness revealed notable distributional patterns. MMLU showed a polarized distribution, with equal proportions of Consistent Bold and Inconsistent Cautious profiles (33.3% each), suggesting that the academic knowledge domain encourages either confident, stable reasoning or variable, consensus-following behavior, with fewer intermediate states. The remaining profiles split evenly between Consistent Cautious and Inconsistent Bold archetypes (16.7% each).

MGSD demonstrated the most balanced distribution across all four archetypes, indicating that stereotype detection elicits a fuller spectrum of behavioral patterns. MentalManip skewed heavily toward cautious profiles (60% combined), reflecting the risk-averse nature of manipulation detection where false positives carry perceived costs. MMLU-Large mirrored MMLU’s patterns but with increased caution, suggesting that greater task difficulty promotes conservative decision-making.

Demographic patterns within archetypes revealed systematic associations that persist across datasets. White profiles predominantly exhibited Consistent Bold behavior in MMLU (70%) but showed more varied patterns in other domains (30% in MGSD), suggesting context-dependent activation of decisiveness. Black profiles demonstrated a tendency toward Consistent Cautious behavior in MMLU (60%) with greater variation elsewhere, while Asian profiles consistently fell into the Inconsistent Cautious category across most datasets (60–70%). Indian profiles showed the highest variability, ranging from 40% Consistent Bold to 60% Consistent Cautious depending on task domain, indicating particularly context-sensitive behavioral patterns.

Normative Value Assessment

The rescue versus extra-error analysis provided crucial insights into the practical implications of consistency patterns. In MMLU, inconsistent profiles demonstrated a small but statistically significant rescue advantage (0.234 vs. 0.219, $\Delta = 0.015, p = 0.035$), with effect size measures confirming practical significance (Cohen’s $h = 0.035$, 95% CI [0.005, 0.069]; Welch’s t-test: $p = 0.037$, Cohen’s $d = 0.559$). This finding suggests that variability in reasoning can occasionally correct errors that

consistent approaches would systematically miss, though the magnitude remains modest.

Other datasets showed negligible differences: MMLU-Large exhibited virtually identical rescue rates (0.178 vs. 0.180, $\Delta = -0.002$, $p = 0.650$), MentalManip showed minimal variation (0.135 vs. 0.134, $\Delta = 0.001$, $p = 0.878$), and MGSD slightly favored consistent profiles (0.119 vs. 0.126, $\Delta = -0.006$, $p = 0.236$). The dataset-specific nature of rescue advantages indicates that the value of consistency depends critically on task characteristics rather than representing a universal principle.

Subgroup analyses revealed further heterogeneity in rescue patterns. Indian profiles in MMLU showed the strongest effect, with inconsistent variants providing substantial rescue advantages ($\Delta = 0.055$, $p = 0.007$, $q = 0.044$ after FDR correction). This ethnicity-specific finding suggests that certain demographic framings may unlock beneficial variability in reasoning. However, most other demographic subgroups showed no significant differences after multiple testing correction, with Middle Eastern profiles in MMLU-Large showing a non-significant trend favoring consistency ($\Delta = -0.013$, $p = 0.127$, $q = 0.382$). Gender and age analyses yielded uniformly null results after FDR correction, though age 20 profiles showed the largest non-significant effects across multiple datasets.

Predictive Attribution and Generalization

Variance decomposition analysis revealed striking differences in how demographic versus other profile traits contribute to behavioral outcomes. For boldness metrics in MMLU, demographic characteristics dominated overwhelmingly: COI showed demographics $R^2 = 0.230$ versus other traits $R^2 = 0.046$ (dominance ratio = 5.03), ATI demonstrated even stronger demographic dominance ($R^2 = 0.510$ vs. 0.024, ratio = 21.08), and CAI followed similar patterns ($R^2 = 0.496$ vs. 0.029, ratio = 17.35). These large dominance ratios suggest that in academic knowledge tasks, demographic framing substantially influences decision-making boldness.

MGSD presented contrasting patterns, with demographic and other traits contributing more equally to variance. COI showed near parity (demographics $R^2 = 0.180$ vs. other $R^2 = 0.191$, ratio = 0.94), while ATI and CAI actually showed slight advantages for non-demographic traits (ratios of 0.84 and 0.83 respectively). This balanced attribution suggests that stereotype detection behaviors emerge from complex interactions between demographic and non-demographic factors rather than simple demographic determinism.

Performance metrics showed similar dataset-specific patterns. Accuracy variance in MGSD was strongly dominated by demographics ($R^2 = 0.439$, dominance = 11.81), while MMLU showed moderate demographic influence ($R^2 = 0.241$, dominance = 5.82). MentalManip and MMLU-Large exhibited weaker demographic effects (dominance ratios of 1.64 and 3.60 respectively), indicating that manipulation detection and com-

plex academic tasks depend less on demographic framing.

Cross-validation analysis exposed substantial optimism in in-sample estimates, with most models showing dramatic performance degradation when tested on held-out folds. The best cross-validated performance occurred for MGSD boldness metrics (ATI: CV $R^2 = 0.281$ vs. in-sample 0.576; CAI: CV $R^2 = 0.334$ vs. in-sample 0.622), still representing approximately 50% shrinkage. Most MMLU metrics showed negative cross-validated R^2 , indicating predictions worse than baseline means, while MentalManip consistently yielded negative values across all metrics. This poor generalization suggests that identified demographic patterns may be largely sample-specific rather than reflecting stable underlying relationships.

Analysis of specific demographic predictors revealed nuanced patterns. Latino profiles emerged as the strongest positive predictors of boldness in MMLU (ATI $\beta = 0.139$, CAI $\beta = 0.133$), while Indian profiles consistently predicted higher rescue rates across datasets ($\beta = 0.022$). Asian profiles often negatively predicted accuracy (β ranging from -0.009 to -0.008), though effect sizes remained small. Age effects showed dataset-dependent patterns, with age 55 consistently predicting higher boldness in MGSD while age 45 showed mixed effects across domains.

Efficiency Considerations

When computational costs were incorporated into the analysis, the marginal benefits of demographic profiling diminished substantially. Single well-chosen profiles frequently matched or exceeded ensemble performance on a per-token basis, as the overhead of running multiple profiles rarely justified marginal accuracy gains. This efficiency analysis suggests that in practical deployments, the computational cost of demographic diversity may outweigh its benefits, particularly given the weak and inconsistent effects observed.

5.4 Discussion

The Consistency-Boldness Trade-off

Our findings reveal that the relationship between consistency and boldness represents a fundamental axis of variation in model behavior, though its manifestation depends critically on task characteristics. The negative correlations observed in MMLU and MGSD support our theoretical framework that stable profiles tend toward bold, non-consensus decisions. This challenges intuitive assumptions that bold behavior necessarily implies erratic or unstable reasoning. Instead, our results suggest that confidence in departing from consensus may actually reflect a form of reasoning stability—a consistent willingness to trust internal judgment over collective wisdom.

The positive association between boldness and accuracy in MMLU indicates that strategic deviation from consensus can improve performance, particularly in domains where the consensus reflects mediocrity rather than excellence. This finding has important implications for ensemble methods and democratic voting schemes, suggesting that uniform weighting of perspectives may actually degrade performance by overvaluing average opinions. However, the negative correlation in MentalManip provides an important counterpoint, demonstrating that consensus-following remains valuable in domains where collective judgment provides useful signal about complex social dynamics.

Demographic Attribution and Its Limits

The variance decomposition results reveal a complex landscape where demographic effects vary dramatically across tasks and metrics. The strong demographic dominance for MMLU boldness metrics suggests that academic knowledge tasks may be particularly susceptible to demographic framing effects, possibly because such framing activates different knowledge domains or reasoning styles learned during training. However, the more balanced attribution in MGSD and the poor cross-validation performance across all datasets caution against overinterpreting these patterns.

The systematic failure of demographic predictors to generalize out-of-sample represents perhaps our most important finding. While in-sample analyses suggest substantial demographic effects, the negative cross-validated R^2 values indicate that these patterns largely reflect overfitting to specific dataset characteristics rather than stable demographic influences on reasoning. This finding fundamentally challenges approaches that attempt to optimize demographic profiles for improved performance, suggesting that such optimization captures noise rather than signal.

Practical Implications for Deployment

The modest rescue advantages observed for inconsistent profiles in MMLU, while statistically significant, must be weighed against their practical costs. An improvement of 1.5 percentage points in rescue rate, particularly when not accompanied by systematic benefits in other metrics, provides weak justification for the complexity of implementing profile-based systems. The heterogeneity of effects across datasets further complicates deployment decisions, as strategies optimal for one domain may prove counterproductive in another.

The subgroup analysis revealing stronger effects for specific demographic combinations (particularly Indian profiles in MMLU) raises important questions about fairness and representation. If certain demographic framings unlock beneficial reasoning patterns, should systems leverage these advantages? The answer depends partly on whether these effects reflect genuine differences in reasoning quality or merely artifacts of training data distributions. Our inability to predict which combinations will prove beneficial suggests that demographic optimization re-

mains more art than science.

Methodological Insights and Recommendations

Several methodological insights emerge from our analysis. First, the gap between in-sample and cross-validated performance underscores the critical importance of proper validation in high-dimensional settings. With 60 profiles and multiple outcome metrics, the risk of discovering spurious associations is substantial, and our FDR correction eliminated most initially significant findings. Future work should prioritize pre-registered analyses and held-out test sets to avoid overfitting.

Second, the equal weighting of profiles in our leave-one-out consensus approach, while principled, may not reflect optimal aggregation strategies. The finding that certain profiles consistently outperform consensus suggests that weighted averaging based on profile characteristics or past performance might yield better results. However, our cross-validation results warn that such weighting schemes may not generalize beyond training data.

Third, the wide confidence intervals on dominance ratios highlight the uncertainty inherent in variance attribution with limited samples. While point estimates suggest strong demographic effects in some domains, the statistical uncertainty around these estimates recommends caution in drawing strong conclusions about the relative importance of demographic versus other factors.

Toward a More Nuanced Understanding

Our results argue for moving beyond simple demographic categorization toward understanding the mechanisms through which different in-context learning strategies influence model behavior. The consistency-boldness framework provides a more informative lens than demographic labels, capturing behavioral patterns that transcend surface-level categories. The finding that demographic traits show statistical associations but poor generalization suggests that these associations may be epiphenomenal correlates rather than causes of behavioral variation.

Future research should investigate the underlying factors that drive consistency and boldness rather than optimizing demographic combinations. Understanding why certain prompts elicit stable versus variable behavior, and why some encourage deviation from consensus, could yield more robust and generalizable improvements than demographic profiling. The task-specific nature of our findings suggests that optimal strategies may need to be discovered empirically for each domain rather than derived from general principles.

Conclusion

This investigation of demographic profiling effects on model behavior yields a nuanced picture that challenges both optimistic claims about diversity benefits and pessimistic concerns about demographic bias. While

demographic characteristics show statistical associations with reasoning patterns, these effects are generally small, task-specific, and fail to generalize beyond training samples. The consistency-boldness framework emerges as a more productive lens for understanding behavioral variation, though its practical utility remains limited.

Given the modest benefits, poor generalization, and computational costs associated with demographic profiling, we recommend that practitioners focus on simpler, more robust approaches. Single-profile strategies, particularly using well-validated prompts, often match or exceed the performance of complex demographic ensembles while requiring fraction of the computational resources. Where behavioral diversity is desired, it may be better achieved through explicit prompt engineering rather than demographic role-playing.

The field would benefit from shifting focus from demographic optimization toward understanding the fundamental mechanisms that drive model behavior across different in-context learning strategies. Our results suggest that the interesting variations in model behavior have less to do with surface-level demographic categories and more to do with how different prompts activate or suppress particular reasoning patterns learned during training. Understanding these deeper patterns promises more robust and generalizable improvements than continued refinement of demographic profiling approaches.

Chapter 6

Discussion and Conclusion

This chapter synthesizes findings from three interconnected studies examining large language model performance, demographic effects, and behavioral patterns under real-world constraints. Through systematic benchmarking, profile-based evaluation, and stability analysis, we develop both diagnostic insights and prescriptive guidance for practical deployment.

6.1 Conclusion

Efficiency in In-Context Learning Strategies

Our benchmarking study challenges the prevailing assumption that sophisticated in-context learning strategies necessarily yield superior results. The dominance of few-shot in-context learning with definitions across diverse tasks demonstrates that simplicity often trumps complexity when computational resources are constrained. While Chain-of-Thought and Tree-of-Thought reasoning have shown promise in specific domains reported in literature, our comprehensive evaluation reveals their limitations: inconsistent accuracy gains that rarely justify their substantially higher token costs.

This finding has profound implications for deployment decisions. Organizations implementing LLMs must balance performance aspirations against operational realities. Our results indicate that for most classification tasks, the optimal strategy involves few-shot prompting with task definitions, achieving approximately 95% of the best possible accuracy while consuming less than half the tokens required by multi-stage reasoning approaches. The marginal gains from complex strategies justify rarely the increase in computational consumption.

The consistent superiority of GPT-4.1-mini across all evaluation dimensions further simplifies deployment decisions. Its combination of high accuracy (67.16% mean) and low variance ($\sigma = 7.61$) across diverse tasks suggests robust internal representations that generalize well beyond specific training distributions. This stability proves particularly valuable in production environments where predictable behavior matters as much as raw performance. The performance gap between GPT-4.1-mini and lower-tier models on challenging datasets indicates fundamental capability differences that cannot be bridged through prompt optimization alone.

Fairness Beyond Demographics

The negligible impact of demographic framing on model performance fundamentally challenges how we conceptualize and evaluate fairness in AI systems. Across four diverse datasets, demographic variables explained less than 3% of performance variance, with the largest observed gap between any demographic groups remaining under 2.8%. These effects pale in comparison to the variations attributable to content characteristics within the same datasets.

This finding redirects our attention from demographic debiasing toward more fundamental challenges in task design and evaluation. The model’s differential performance across stereotype categories in MGSD, which range from 63% accuracy on gender stereotypes to 75% on professional stereotypes, reveals that ambiguity in task definition poses far greater challenges than demographic bias. Rather than exhibiting systematic prejudice, the model appears to faithfully reflect the inherent noise and disagreement present in human-generated training data.

The occasional detection of subtle patterns through permutation testing, while statistically significant, lacks practical importance. These effects emerge sporadically, fail to replicate across datasets, and disappear entirely under cross-validation. This pattern suggests that much of what appears as demographic influence in small-sample studies may actually represent statistical artifacts rather than genuine behavioral differences. The field’s focus on demographic fairness, while important in principle, may be diverting attention from more pressing issues of task clarity, annotation quality, and fundamental model capabilities.

LLMs Behaviour

The consistency-boldness framework reveals behavioral patterns that transcend demographic categories, offering a more nuanced understanding of model variation. The negative correlation between consistency and boldness in academic knowledge tasks ($r = -0.316$ for MMLU) suggests that stable reasoning patterns often accompany willingness to deviate from consensus. This relationship inverts our intuitive understanding: boldness emerges not from erratic behavior but from consistent confidence in internal judgment.

The positive correlation between boldness and accuracy in MMLU ($r = 0.528$) carries important implications for ensemble methods. When consensus reflects mediocrity rather than excellence, strategic deviation improves performance. This finding challenges democratic aggregation approaches that weight all perspectives equally, suggesting that identifying and amplifying bold, accurate voices could yield better results than simple majority voting. However, the negative correlation in MentalManip ($r = -0.282$) provides crucial context: in domains involving social judgment, consensus-following remains valuable.

The archetype analysis revealing demographic associations with behavioral patterns suggests that demographic framing may influence reasoning style even when it doesn’t affect accuracy, such as White profiles trending toward Consistent Bold behavior in MMLU. Yet these associations prove unstable and task-dependent, varying dramatically across

datasets and failing to generalize out-of-sample. The systematic failure of demographic predictors under cross-validation, with most models showing negative R^2 values, indicates that these patterns largely reflect overfitting rather than stable behavioral tendencies.

6.1.1 Research Implications

Rethinking Evaluation Priorities

Our findings argue for fundamental shifts in how we evaluate and optimize language models. The current emphasis on demographic fairness, while ethically important, may be misallocating research effort given the minimal practical impact of demographic variables on performance. Instead, research should prioritize understanding and addressing the content-specific challenges that account for the vast majority of performance variation.

The discrepancy between in-sample and out-of-sample performance in our predictive models underscores the dangers of over-interpretation in high-dimensional settings. With 60 profiles and multiple outcome metrics, the probability of discovering spurious associations approaches certainty without rigorous validation. Future work must adopt more conservative analytical approaches, including pre-registered analyses, held-out test sets, and aggressive correction for multiple comparisons.

The consistency-boldness framework offers a more productive lens than demographic categorization for understanding model behavior. Rather than optimizing demographic combinations research should investigate the mechanisms that drive consistent versus variable behavior and consensus-following versus deviation. Understanding these deeper patterns promises more robust and generalizable improvements than surface-level demographic adjustments.

Practical Deployment Strategies

For practitioners, our results provide clear guidance: simple, well-validated in-context learning strategies typically match or exceed complex alternatives while requiring fraction of the computational resources. The optimal approach for most applications involves few-shot prompting with task definitions using GPT-4.1-mini or comparable high-performance models. Attempts to optimize performance through demographic profiling or complex reasoning chains are unlikely to yield returns justifying their costs.

Where behavioral diversity is desired, explicit prompt engineering proves more effective than demographic role-playing. Direct instructions to consider alternative viewpoints, challenge assumptions, or explore edge cases produce more reliable variation than asking models to embody different demographic personas.

The modest rescue advantages observed for inconsistent profiles (1.5% in MMLU) provide insufficient justification for implementing complex ensemble systems. Single-profile approaches using validated prompts typically achieve 95-98% of ensemble performance while eliminating coordination overhead and reducing computational costs by orders of magnitude.

Only in specialized applications where even marginal improvements justify substantial complexity might profile-based approaches warrant consideration.

Methodological Contributions

Beyond specific findings, this work contributes methodological frameworks applicable to future research. The three-tier statistical approach provides a template for detecting subtle effects while maintaining appropriate skepticism about their practical significance by combining traditional parametric tests, effect size estimation, and non-parametric permutation testing. This multi-pronged strategy proves particularly valuable when evaluating high-dimensional systems where spurious correlations abound.

The consistency-boldness measurement framework offers new tools for characterizing model behavior beyond simple accuracy metrics. By quantifying the relationship between stability and decision confidence, researchers can identify behavioral patterns that traditional evaluations miss. The leave-one-out consensus approach for measuring boldness, while computationally intensive, reveals important variations in how models balance individual judgment against collective wisdom.

The variance decomposition methodology, separating demographic from non-demographic contributions, helps researchers understand the sources of behavioral variation in complex systems. While our specific findings suggest minimal demographic influence, the approach itself transfers to other domains where understanding variance sources matters for system design and evaluation.

6.1.2 Limitations and Future Directions

Several limitations constrain the generalizability of our findings. Our focus on classification tasks may not capture effects that emerge in generation or other open-ended applications where demographic framing might have larger impacts. The use of fixed dataset samples and zero temperature, while ensuring reproducibility, may underestimate the full range of model behaviors observable in production settings.

The demographic profiles examined, while diverse, cannot capture the full complexity of human identity and perspective. Intersectional effects, cultural nuances, and individual variations beyond our demographic categories may influence model behavior in ways our framework cannot detect. Future work should explore richer representations of perspective and identity, though our results suggest that even elaborate demographic modeling may yield minimal practical benefits.

The rapid evolution of model capabilities means our specific performance numbers will quickly become obsolete. However, the patterns we identify likely reflect fundamental characteristics that will persist across model generations.

Future research should investigate whether our findings extend to other model families, particularly open-source alternatives that may exhibit different behavioral patterns. The mechanisms underlying consistency and boldness warrant deeper investigation, potentially through in-

interpretability methods that can identify which internal representations drive these behavioral dimensions. Understanding why certain prompts elicit stable versus variable behavior could lead to more principled approaches to prompt engineering.

Concluding Remarks

This comprehensive investigation of large language model behavior under real-world constraints yields a nuanced picture that challenges both optimistic and pessimistic narratives about AI capabilities and fairness. While demographic characteristics show occasional statistical associations with model behavior, these effects prove small, unstable, and practically negligible compared to content-driven variation. The consistency-boldness framework provides more insight into behavioral patterns than demographic categorization, though even these dimensions show limited practical utility for performance optimization.

The dominance of simple in-context learning strategies over complex alternatives, demonstrated across diverse tasks and models, argues for pragmatism in deployment decisions. Organizations should resist the allure of sophisticated techniques that promise marginal gains while imposing substantial costs. Similarly, concerns about demographic bias, while ethically important, should not overshadow more fundamental challenges in task design, annotation quality, and model capability development.

Our findings ultimately argue for redirecting research effort from surface-level optimization toward understanding fundamental mechanisms of model behavior. Rather than engineering increasingly complex demographic profiles or reasoning chains, the field would benefit from developing better benchmarks, clearer task definitions, and more robust evaluation frameworks. The path toward more capable and reliable AI systems lies not in elaborate in-context learning gymnastics but in addressing the core challenges of ambiguity, consistency, and generalization that our studies reveal.

As large language models become increasingly embedded in high-stakes applications, the need for rigorous, pragmatic evaluation grows ever more critical. This work provides both methodological frameworks and empirical findings to guide that evaluation, offering a foundation for more informed decisions about when, how, and why to deploy these powerful but imperfect systems. The ultimate goal remains not perfect fairness or optimal performance but practical utility: developing and deploying AI systems that reliably serve human needs within realistic constraints.

Appendix A

Resources

A.1 Code

The supporting code for this thesis is implemented in the following repository: https://github.com/raphaelyana/normative_reasoning_and_stereotypes

Appendix B

Study 2: Additional Contents

B.1 Empirical Group-Level Accuracy

For two given trait groups G_1 and G_2 , each being a set of profiles, let $\hat{a}_G$ denote the empirical group-level accuracy:

$$\hat{a}_G = \frac{1}{|G|} \sum_{p \in G} \hat{p}_p \quad (\text{B.1.1})$$

It corresponds to the mean of accuracy per profile within a given group for all profiles in the group G .

We associate this metric to Cohen's h distance to report effect size independent of scale, calculated as:

$$h = 2 \arcsin \sqrt{\hat{a}_{G_1}} - 2 \arcsin \sqrt{\hat{a}_{G_2}} \quad (\text{B.1.2})$$

B.2 Error Directionality

This is a formal mathematical definition of the error directionality we defined in this thesis.

Mathematical Formulation To inspect performance of a given profile compared to a fixed baseline in more details, we introduce *Error correction* and *Extra-Errors* as additional metrics. Using our notations defined in Chapter 2, for a baseline $p_0 \in \mathcal{P}$ and a given set of evaluated samples $\mathcal{J}$, we define the following two sets based on the baselines results and the samples evaluated:

$$\mathcal{E}_{p_0} = \{i \in \mathcal{J} : Y_{i,p_0} = 0\} \quad (\text{B.2.1})$$

and:

$$\mathcal{C}_{p_0} = \{i \in \mathcal{J} : Y_{i,p_0} = 1\} \quad (\text{B.2.2})$$

representing the sets of incorrect and correct predictions made by the baseline p_0 (respectively). We then define the following counts based on the paired outcomes $(Y_{i,p_0}, Y_{i,p}) \in \{0, 1\}^2$ of the baseline profile p_0 and a given role-played profile p :

$$C^+(p; p_0) = \sum_{i \in \mathcal{J}} \mathbf{1}[Y_{i,p_0} = 0, Y_{i,p} = 1] \quad (\text{B.2.3})$$

and:

$$C^-(p; p_0) = \sum_{i \in \mathcal{I}} \mathbf{1}[Y_{i,p_0} = 1, Y_{i,p} = 0] \quad (\text{B.2.4})$$

where $\mathbf{1}$ is an indicator function¹ [1], C^+ represents the Error Correction, *i.e.* when the baseline prediction is *incorrect* and the profile prediction is *correct*, and C^- the Extra-Error, *i.e.* when the baseline prediction is *correct* and the profile is *incorrect*. These counts permit to define the Error Correction rate and the Extra-Error rate, EC and EE respectively, as following:

$$\text{EC}(p|p_0) = \frac{C^+(p; p_0)}{|\mathcal{E}_{p_0}|} \quad (\text{B.2.5})$$

And:

$$\text{EE}(p|p_0) = \frac{C^-(p; p_0)}{|\mathcal{C}_{p_0}|} \quad (\text{B.2.6})$$

The net accuracy change relative to the baseline can be defined similarly as:

$$\Delta_{\text{acc}}(p; p_0) = \frac{C^+(p; p_0) - C^-(p; p_0)}{|\mathcal{I}|} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} (Y_{i,p} - Y_{i,p_0}) \quad (\text{B.2.7})$$

Associated Statistical Tools To complete these metrics and strengthen the analysis, we display McNemar's test to assess directional asymmetry, defined in Chapter 2. McNemar's test will test if the introduction of error correction and extra-errors by a profile compared to the fixed baseline are symmetric, *i.e.* if the following hypothesis is true:

$$H_0 : \mathbb{P}(Y_{i,p_0} = 0, Y_{i,p} = 1) = \mathbb{P}(Y_{i,p_0} = 1, Y_{i,p} = 0) \quad (\text{B.2.8})$$

We let $\mathbb{C} = C^+(p; p_0) + C^-(p; p_0)$ to denote the set of counts, therefore the two-sided p -value can be obtained from:

$$X \sim \text{Binom}\left(\mathbb{C}, \frac{1}{2}\right)$$

as:

$$p_{\text{exact}} = 2\text{Pr}[X \leq \min(C^+, C^-)] \quad (\text{B.2.9})$$

We also report the corresponding odds ratio:

$$\hat{\theta} = \frac{C^+}{C^-} \quad (\text{B.2.10})$$

and the associated log-normal 95% confidence interval, called Wald confidence interval:

$$\text{CI}_{\text{Wald}} = \exp \left\{ \log \hat{\theta} \pm 1.96 \sqrt{\frac{1}{C^+} + \frac{1}{C^-}} \right\} \quad (\text{B.2.11})$$

the Haldane-Anscombe correction :

¹ $\mathbf{1}[X] = 1$ if X is True, 0 if X is false.

$$\hat{\theta}_{\text{corrected}} = \frac{C^+ + 0.5}{C^- + 0.5} \quad (\text{B.2.12})$$

and its corresponding 95% log-normal CI:

$$\text{CI}_{\text{Wald,corrected}} = \exp \left\{ \log \hat{\theta}_{\text{corrected}} \pm 1.96 \sqrt{\frac{1}{C^+ + 0.5} + \frac{1}{C^- + 0.5}} \right\} \quad (\text{B.2.13})$$

B.3 High-disagreement evaluation

We introduce a descriptive metric to analyse qualitatively the obtained results given a sample: the disagreement between different profiles.

Let $\mathcal{K}$ be the set of valid labels. For each item i and each $k \in \mathcal{K}$, let n_k denote the number of profiles predicting label k . Thus, for P profiles, $\sum_k n_k = P$. This metric is computed independently for each item, and reflects the agreement across profiles on that item.

Given P profiles, the disagreement score is:

$$D = 1 - \max_k \frac{n_k}{P} \quad (\text{B.3.1})$$

representing the fraction of profiles predicting a different label from the majority's prediction. A smaller value for D indicates a strong consensus, and permits to evaluate difficult items from the dataset.

We associate the D score to Shannon Entropy H of the empirical prediction distribution:

$$H = - \sum_{k \in \mathcal{K}} \frac{n_k}{P} \log_2 \left(\frac{n_k}{P} \right) \quad (\text{B.3.2})$$

It permits to evaluate the spread of predictions across the labels.

B.4 Additional Results

We provide in this section additional interesting results that are worth keeping.

Level 2: MGSD

This table on MGSD dataset shows no interaction term has significance since all OR are near 1, suggesting limited evidence of gender-specific departures by ethnicity.

Table B.1: Pairwise ethnicity contrasts from GLM (BHFDR q shown).

Factor	Level _{<i>i</i>}	Level _{<i>j</i>}	Logit diff	OR ratio	SE	Stat	p	q
ethnicity	black	indian	-0.1045	0.9008	0.0390	7.1903	0.0076	0.0813
ethnicity	black	middle_eastern	-0.0966	0.9079	0.0378	6.5397	0.0108	0.0813
ethnicity	indian	latine	+0.0606	1.0624	0.0311	3.8028	0.0517	0.2020
ethnicity	asian	indian	-0.0625	0.9394	0.0323	3.7345	0.0539	0.2020
ethnicity	white	black	+0.0770	1.0800	0.0462	2.7703	0.0967	0.2648
ethnicity	asian	middle_eastern	-0.0546	0.9469	0.0337	2.6238	0.1059	0.2648
ethnicity	latine	middle_eastern	-0.0527	0.9487	0.0366	2.0751	0.1503	0.3222
ethnicity	black	latine	-0.0440	0.9570	0.0332	1.7514	0.1863	0.3493
ethnicity	asian	black	+0.0420	1.0429	0.0342	1.5117	0.2195	0.3658
ethnicity	white	latine	+0.0330	1.0336	0.0402	0.6759	0.4114	0.6171
ethnicity	white	asian	+0.0349	1.0356	0.0542	0.4161	0.5192	0.6936
ethnicity	white	indian	-0.0275	0.9728	0.0466	0.3492	0.5548	0.6936
ethnicity	white	middle_eastern	-0.0196	0.9806	0.0546	0.1296	0.7191	0.8297
ethnicity	indian	middle_eastern	+0.0079	1.0079	0.0334	0.0558	0.8133	0.8714
ethnicity	asian	latine	-0.0019	0.9981	0.0353	0.0030	0.9565	0.9565

Bibliography

- [1] George Casella and Roger Berger. *Statistical inference*. Chapman and Hall/CRC, 2024.
- [2] Margherita Grandini, Enrico Bagli, and Giorgio Visani. “Metrics for Multi-Class Classification: an Overview”. In: *arXiv preprint arXiv:2008.05756* (Aug. 2020). URL: <https://doi.org/10.48550/arXiv.2008.05756>.
- [3] Jan Broersen et al. “Normative Reasoning and Consequence”. In: *Normative Multi-Agent Systems*. Vol. 4. Dagstuhl Follow-Ups. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2013, pp. 33–70. URL: <https://doi.org/10.4230/DFU.Vol4.12111.33>.
- [4] Tingxu Han et al. “Token-Budget-Aware LLM Reasoning”. In: *arXiv preprint arXiv:2412.18547* (2024). Version 5, last revised 2 Jun 2025. arXiv: 2412.18547 [cs.CL]. URL: <https://arxiv.org/abs/2412.18547>.
- [5] Junlin Wang et al. “Reasoning in Token Economies: Budget-Aware Evaluation of LLM Reasoning Strategies”. In: *arXiv preprint arXiv:2406.06461* (2024). Version 3, last revised 15 Jun 2024. arXiv: 2406.06461 [cs.CL]. URL: <https://arxiv.org/abs/2406.06461>.
- [6] Microsoft Learn. *A list of metrics for evaluating LLMgenerated content*. Accessed 2025-09-22. June 2024. URL: <https://learn.microsoft.com/en-us/ai/playbook/technology-guidance/generative-ai/working-with-llms/evaluation/list-of-eval-metrics>.
- [7] Conor Bronsdon. *7 Key LLM Metrics to Enhance AI Reliability*. Accessed 2025-09-22. Mar. 2025. URL: <https://galileo.ai/blog/llm-performance-metrics>.
- [8] Jason Wei et al. “Finetuned language models are zero-shot learners”. In: *arXiv preprint arXiv:2109.01652* (2021).
- [9] Hugo Touvron et al. “Llama: Open and efficient foundation language models”. In: *arXiv preprint arXiv:2302.13971* (2023).
- [10] Tom B. Brown et al. “Language Models are Few-Shot Learners”. In: *arXiv preprint arXiv:2005.14165* (July 2020). Version 4, last revised 22 Jul 2020. URL: <https://doi.org/10.48550/arXiv.2005.14165>.
- [11] Jiawei Gu et al. “A Survey on LLM-as-a-Judge”. In: *arXiv preprint arXiv:2411.15594* (Nov. 2024). Version 5, last revised 9 Mar 2025. URL: <https://doi.org/10.48550/arXiv.2411.15594>.

- [12] Lianmin Zheng et al. “Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena”. In: *arXiv preprint arXiv:2306.05685* (Dec. 2023). Version 4, last revised 24 Dec 2023. NeurIPS 2023 Datasets and Benchmarks Track. URL: <https://doi.org/10.48550/arXiv.2306.05685>.
- [13] Songze Li et al. “LLMs Cannot Reliably Judge (Yet?): A Comprehensive Assessment on the Robustness of LLM-as-a-Judge”. In: *arXiv preprint arXiv:2506.09443* (June 2025). Version 1, submitted on 11 Jun 2025. URL: <https://doi.org/10.48550/arXiv.2506.09443>.
- [14] Chengwei Qin et al. “In-Context Learning with Iterative Demonstration Selection”. In: *arXiv preprint arXiv:2310.09881* (Dec. 2023). Version 4, last revised 30 Dec 2024. URL: <https://doi.org/10.48550/arXiv.2310.09881>.
- [15] Branislav Pecher et al. “Automatic combination of sample selection strategies for few-shot learning”. In: *arXiv preprint arXiv:2402.03038* (2024).
- [16] Hyunbin Jin et al. ““Well, Keep Thinking”: Enhancing LLM Reasoning with Adaptive Injection Decoding”. In: *arXiv preprint arXiv:2503.10167* (Mar. 2025). Version 2, last revised 18 Mar 2025. URL: <https://doi.org/10.48550/arXiv.2503.10167>.
- [17] Jia He et al. “Does Prompt Formatting Have Any Impact on LLM Performance?” In: *arXiv preprint arXiv:2411.10541* (Nov. 2024). Submitted to NAACL 2025. URL: <https://doi.org/10.48550/arXiv.2411.10541>.
- [18] Isabel O Gallegos et al. “Bias and fairness in large language models: A survey”. In: *Computational Linguistics* 50.3 (2024), pp. 1097–1179.
- [19] Shashank Gupta et al. “Bias runs deep: Implicit reasoning biases in persona-assigned llms”. In: *arXiv preprint arXiv:2311.04892* (2023).
- [20] Qihao Zhao et al. “Mmlu-cf: A contamination-free multi-task language understanding benchmark”. In: *arXiv preprint arXiv:2412.15194* (2024).
- [21] Yuxin Wang et al. “MentalManip: A dataset for fine-grained analysis of mental manipulation in conversations”. In: *arXiv preprint arXiv:2405.16584* (2024).
- [22] Zekun Wu et al. “Stereotype Detection in LLMs: A Multiclass, Explainable, and Benchmark-Driven Approach”. In: *arXiv preprint arXiv:2404.01768* (2024).
- [23] Theo King et al. “Hearts: A holistic framework for explainable, sustainable and robust text stereotype detection”. In: *arXiv preprint arXiv:2409.11579* (2024).
- [24] Ziqiao Ma et al. “Towards a holistic landscape of situated theory of mind in large language models”. In: *arXiv preprint arXiv:2310.19619* (2023).

- [25] Simon Baron-Cohen, Alan M Leslie, and Uta Frith. “Does the autistic child have a theory of mind?” In: *Cognition* 21.1 (1985), pp. 37–46.
- [26] Matthew Riemer et al. “Position: Theory of Mind Benchmarks are Broken for Large Language Models”. In: *arXiv preprint arXiv:2412.19726* (2024).
- [27] Qiaosi Wang et al. “Rethinking theory of mind benchmarks for llms: Towards a user-centered perspective”. In: *arXiv preprint arXiv:2504.10839* (2025).
- [28] Cheng Xu et al. “DCR: Quantifying Data Contamination in LLMs Evaluation”. In: *arXiv preprint arXiv:2507.11405* (2025).
- [29] Yujian Fu et al. “Does data contamination detection work (well) for llms? a survey and evaluation on detection assumptions”. In: *arXiv preprint arXiv:2410.18966* (2024).
- [30] Yaoyang Liu et al. “A comprehensive taxonomy of prompt engineering techniques for large language models”. In: *Frontiers of Computer Science* (2025). doi 10 (2025).
- [31] EG Santana Jr et al. “Which Prompting Technique Should I Use? An Empirical Investigation of Prompting Techniques for Software Engineering Tasks”. In: *arXiv preprint arXiv:2506.05614* (2025).
- [32] Negar Arabzadeh and Charles LA Clarke. “A human-ai comparative analysis of prompt sensitivity in llm-based relevance judgment”. In: *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 2025, pp. 2784–2788.
- [33] Rahul K Arora et al. “Healthbench: Evaluating large language models towards improved human health”. In: *arXiv preprint arXiv:2505.08775* (2025).
- [34] Jairo Gudiño-Rosero et al. “Prompt Injection Vulnerability of Consensus Generating Applications in Digital Democracy”. In: *arXiv preprint arXiv:2508.04281* (2025).
- [35] Brennen Hill et al. “Breaking to Build: A Threat Model of Prompt-Based Attacks for Securing LLMs”. In: *arXiv preprint arXiv:2509.04615* (2025).
- [36] Luca Beurer-Kellner et al. “Design patterns for securing llm agents against prompt injections”. In: *arXiv preprint arXiv:2506.08837* (2025).
- [37] Zili Wang et al. “Faster and better llms via latency-aware test-time scaling”. In: *arXiv preprint arXiv:2505.19634* (2025).
- [38] Zili Wang et al. “Faster and better llms via latency-aware test-time scaling”. In: *arXiv preprint arXiv:2505.19634* (2025).
- [39] Xinyue Li et al. “Benchmarking bias in large language models during role-playing”. In: *arXiv preprint arXiv:2411.00585* (2024).
- [40] Adam Karvonen and Samuel Marks. “Robustly Improving LLM Fairness in Realistic Settings via Interpretability”. In: *arXiv preprint arXiv:2506.10922* (2025).

- [41] Zhengyu Hu et al. “Population-Aligned Persona Generation for LLM-based Social Simulation”. In: *arXiv preprint arXiv:2509.10127* (2025).
- [42] Jinman Zhao et al. “Role-play paradox in large Language models: reasoning performance gains and ethical dilemmas”. In: *arXiv preprint arXiv:2409.13979* (2024).
- [43] Bryan Chen Zhengyu Tan and Roy Ka-Wei Lee. “Unmasking Implicit Bias: Evaluating Persona-Prompted LLM Responses in Power-Disparate Social Scenarios”. In: *arXiv preprint arXiv:2503.01532* (2025).
- [44] Emilio Villa-Cueva et al. “MOMENTS: A Comprehensive Multi-modal Benchmark for Theory of Mind”. In: *arXiv preprint arXiv:2507.04415* (2025).
- [45] Yi-Long Lu et al. “Do Theory of Mind Benchmarks Need Explicit Human-like Reasoning in Language Models?” In: *arXiv preprint arXiv:2504.01698* (2025).
- [46] Jian Li et al. “A survey on benchmarks of multimodal large language models”. In: *arXiv preprint arXiv:2408.08632* (2024).
- [47] Shiwen Ni et al. “A survey on large language model benchmarks”. In: *arXiv preprint arXiv:2508.15361* (2025).
- [48] Mahmoud Mohammadi et al. “Evaluation and benchmarking of llm agents: A survey”. In: *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*. 2025, pp. 6129–6139.
- [49] Fanghua Ye et al. “Benchmarking llms via uncertainty quantification”. In: *Advances in Neural Information Processing Systems 37* (2024), pp. 15356–15385.
- [50] Shengwei Xu et al. “Benchmarking LLMs’ Judgments with No Gold Standard”. In: *arXiv preprint arXiv:2411.07127* (2024).
- [51] Haiquan Hu et al. “STEM: Efficient Relative Capability Evaluation of LLMs through Structured Transition Samples”. In: *arXiv preprint arXiv:2508.12096* (2025).
- [52] Wei Ma et al. “Prompt Stability in Code LLMs: Measuring Sensitivity and Robustness Across Model Families”. In: *arXiv preprint arXiv:2509.13680* (2025). Preprint, accessed September 2025. URL: <https://arxiv.org/abs/2509.13680>.
- [53] Federico Errica et al. “What did I do wrong? Quantifying LLMs’ sensitivity and consistency to prompt engineering”. In: *arXiv preprint arXiv:2406.12334* (2024).
- [54] Kwesi Cobbina and Tianyi Zhou. “Where to show Demos in Your Prompt: A Positional Bias of In-Context Learning”. In: *arXiv preprint arXiv:2507.22887* (2025). arXiv: 2507.22887 [cs.CL]. URL: <https://arxiv.org/abs/2507.22887>.
- [55] Joshua Vendrow et al. “Large Language Model Benchmarks Do Not Test Reliability”. In: *Neurips Safe Generative AI Workshop 2024*. 2024. URL: <https://openreview.net/forum?id=XSeN6xZtZ9>.

- [56] Elena Merdjanovska, Ansar Aynetdinov, and Alan Akbik. “NoiseBench: Benchmarking the Impact of Real Label Noise on Named Entity Recognition”. In: *arXiv preprint arXiv:2405.07609* (May 2024). Submitted 13 May 2024, revised version Oct 2024. URL: <https://arxiv.org/abs/2405.07609>.
- [57] Md Rakibul Hasan et al. “UPLME: Uncertainty-Aware Probabilistic Language Modelling for Robust Empathy Regression”. In: *arXiv preprint arXiv:2508.03520* (Aug. 2025). Submitted on 5 Aug 2025. URL: <https://arxiv.org/abs/2508.03520>.
- [58] Siddhartha Devic et al. “From Calibration to Collaboration: LLM Uncertainty Quantification Should Be More Human-Centered”. In: *arXiv preprint arXiv:2506.07461* (2025). arXiv: 2506.07461 [cs.CL]. URL: <https://arxiv.org/abs/2506.07461>.
- [59] Bingchen Wang, Zi-Yu Khoo, and Bryan Kian Hsiang Low. “Prompts to Proxies: Emulating Human Preferences via a Compact LLM Ensemble”. In: *arXiv preprint arXiv:2509.11311* (2025). Preprint of work originally submitted to AAAI 2026. Under revision for resubmission to a machine learning venue. arXiv: 2509.11311 [cs.AI]. URL: <https://arxiv.org/abs/2509.11311>.
- [60] A. Leiva-Araos, B. Gana, H. Allende-Cid, et al. “Large scale summarization using ensemble prompts and in context learning approaches”. In: *Scientific Reports* 15 (2025), p. 10259. URL: <https://doi.org/10.1038/s41598-025-94551-8>.
- [61] Sungjin Park et al. “Ensembling Large Language Models with Process Reward-Guided Tree Search for Better Complex Reasoning”. In: *arXiv preprint arXiv:2412.15797* (2024). arXiv: 2412.15797 [cs.CL]. URL: <https://arxiv.org/abs/2412.15797>.
- [62] Hao Wen et al. “BudgetThinker: Empowering Budget-aware LLM Reasoning with Control Tokens”. In: *arXiv preprint arXiv:2508.17196* (2025). Version 2, last revised 29 Aug 2025. arXiv: 2508.17196 [cs.LG]. URL: <https://arxiv.org/abs/2508.17196>.
- [63] Yichong Huang et al. “Ensemble learning for heterogeneous large language models with deep parallel collaboration”. In: *Advances in Neural Information Processing Systems* 37 (2024), pp. 119838–119860.
- [64] Nathaniel Demchak et al. “Assessing Bias in Metric Models for LLM Open-Ended Generation Bias Benchmarks”. In: *NeurIPS 2024 Workshop on Evaluations for Evaluations (EvalEval)*. 2024. arXiv: 2410.11059 [cs.CL]. URL: <https://arxiv.org/abs/2410.11059>.
- [65] OpenAI. *Hello GPT-4o*. OpenAI website. Accessed 2025-09-22. May 2024. URL: <https://openai.com/index/hello-gpt-4o/>.
- [66] OpenAI. *GPT-4o mini: Advancing Cost-Efficient Intelligence*. OpenAI website. Accessed 2025-09-22. July 2024. URL: <https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/>.

- [67] Anthropic. *Claude 3.7 Sonnet and Claude Code*. Anthropic website. Feb. 2025. URL: <https://www.anthropic.com/news/claude-3-7-sonnet>.
- [68] Open Laboratory. *Mistral Small 3.2 (2506)*. Open Laboratory website. Accessed 2025-09-22. June 2025. URL: https://openlaboratory.ai/models/mistral-small-3_2-24b-instruct-2506/.
- [69] Mistral AI. *Mistral Small 3.1: The best model in its weight class*. Mistral AI website. Accessed 2025-09-22. Mar. 2025. URL: <https://mistral.ai/news/mistral-small-3-1>.
- [70] DeepSeek-AI. “DeepSeek-V3 Technical Report”. In: *arXiv preprint arXiv:2412.19437v1* (2024). Accessed 2025-09-22. URL: <https://arxiv.org/html/2412.19437v1>.